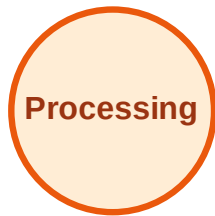


BFS Traversal

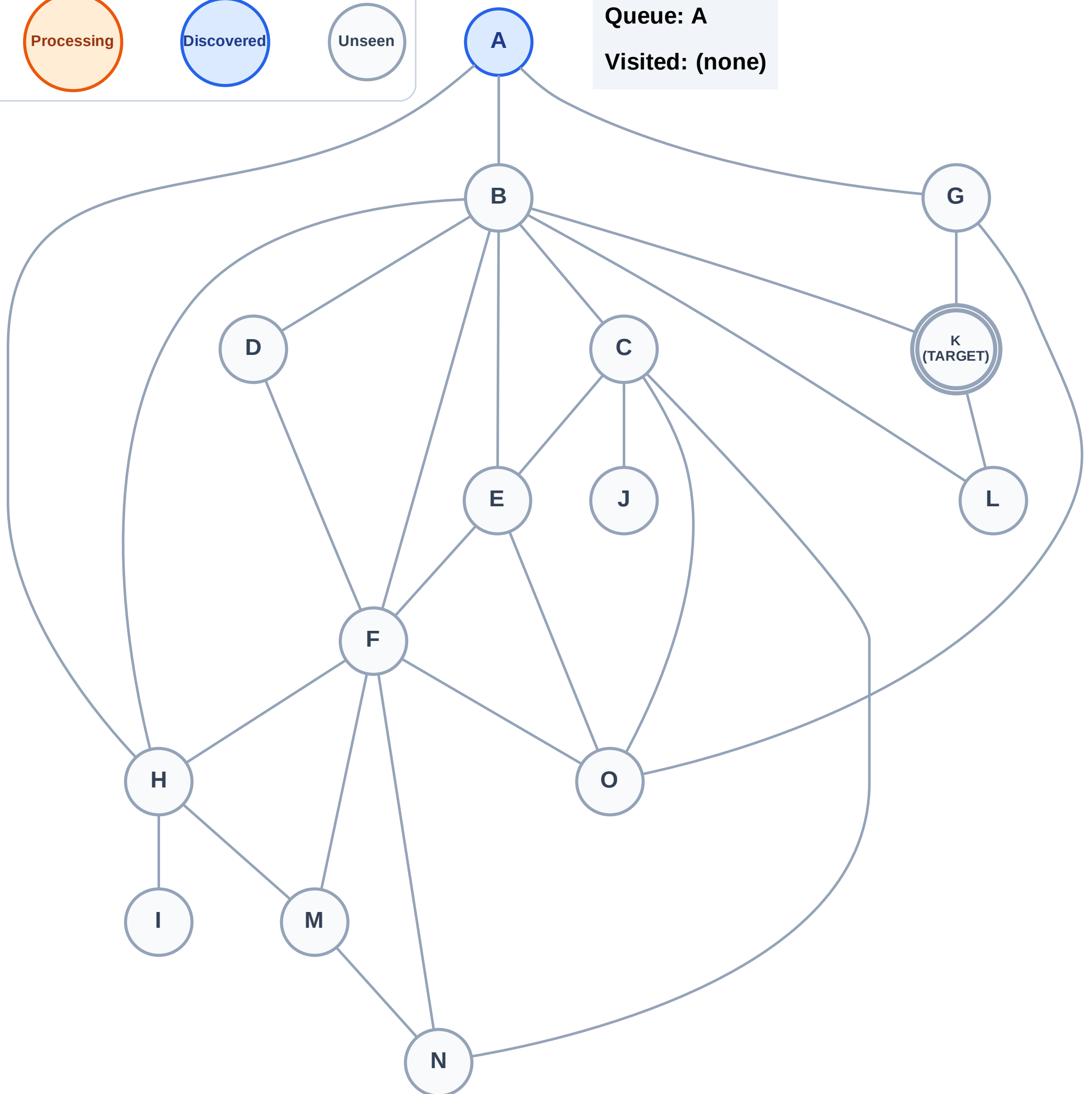
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

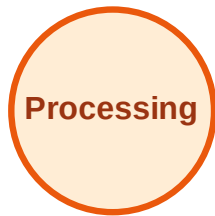
Visited: (none)



BFS Traversal

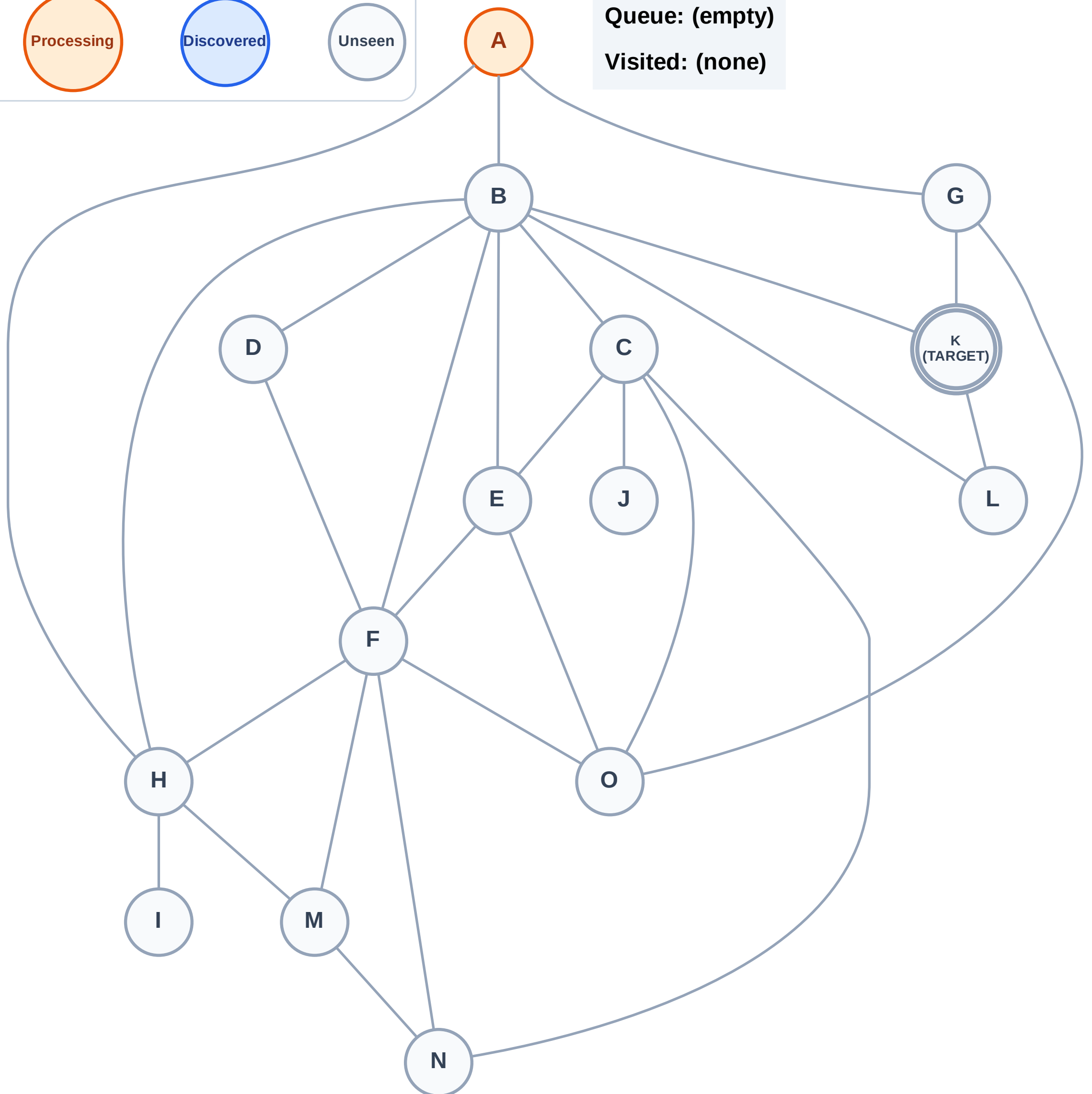
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



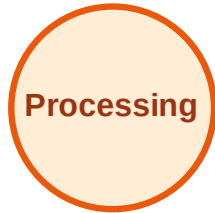
BFS Traversal

Step 3: Discover B, G, H from A

Legend



Complete



Processing



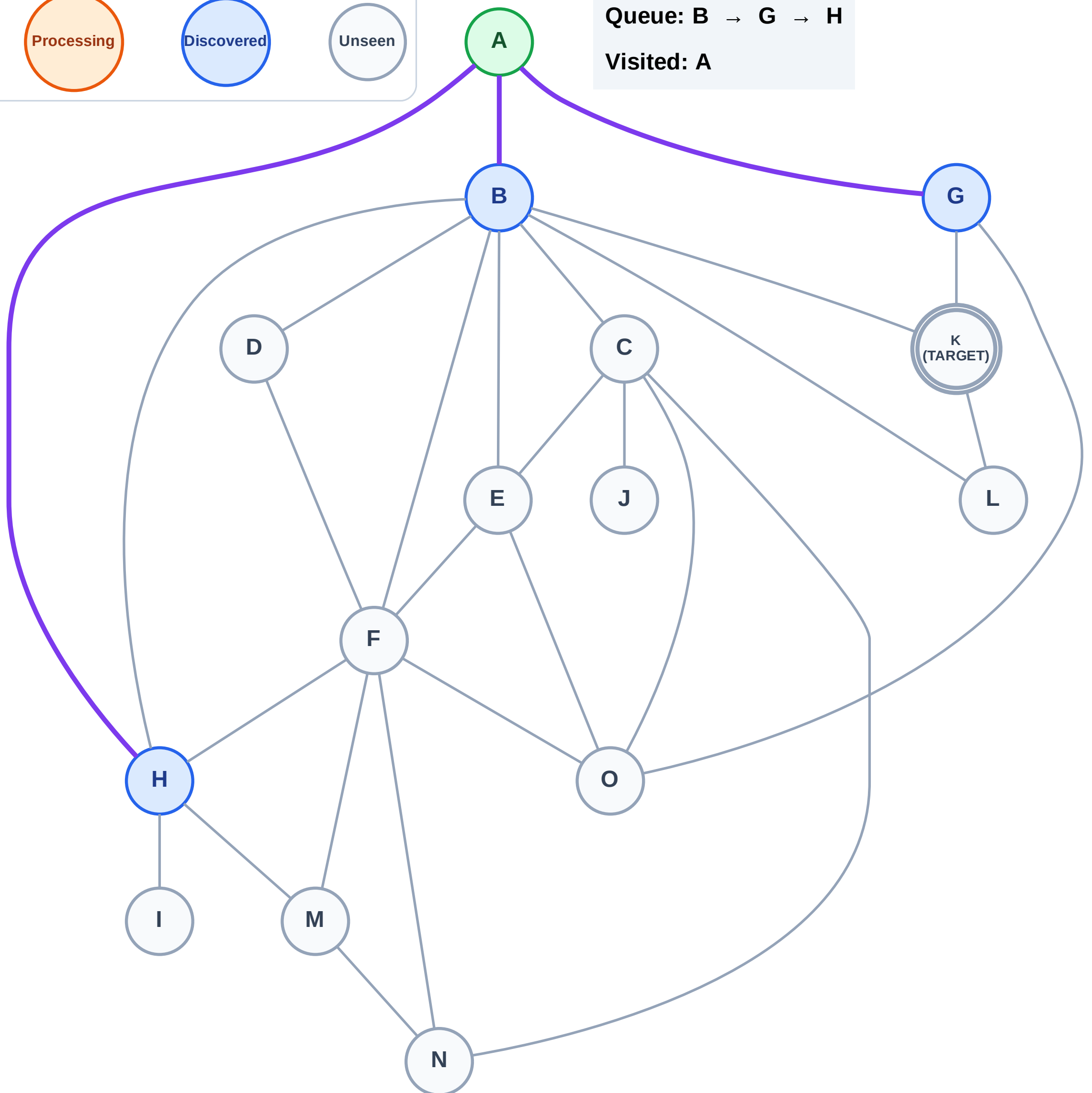
Discovered



Unseen

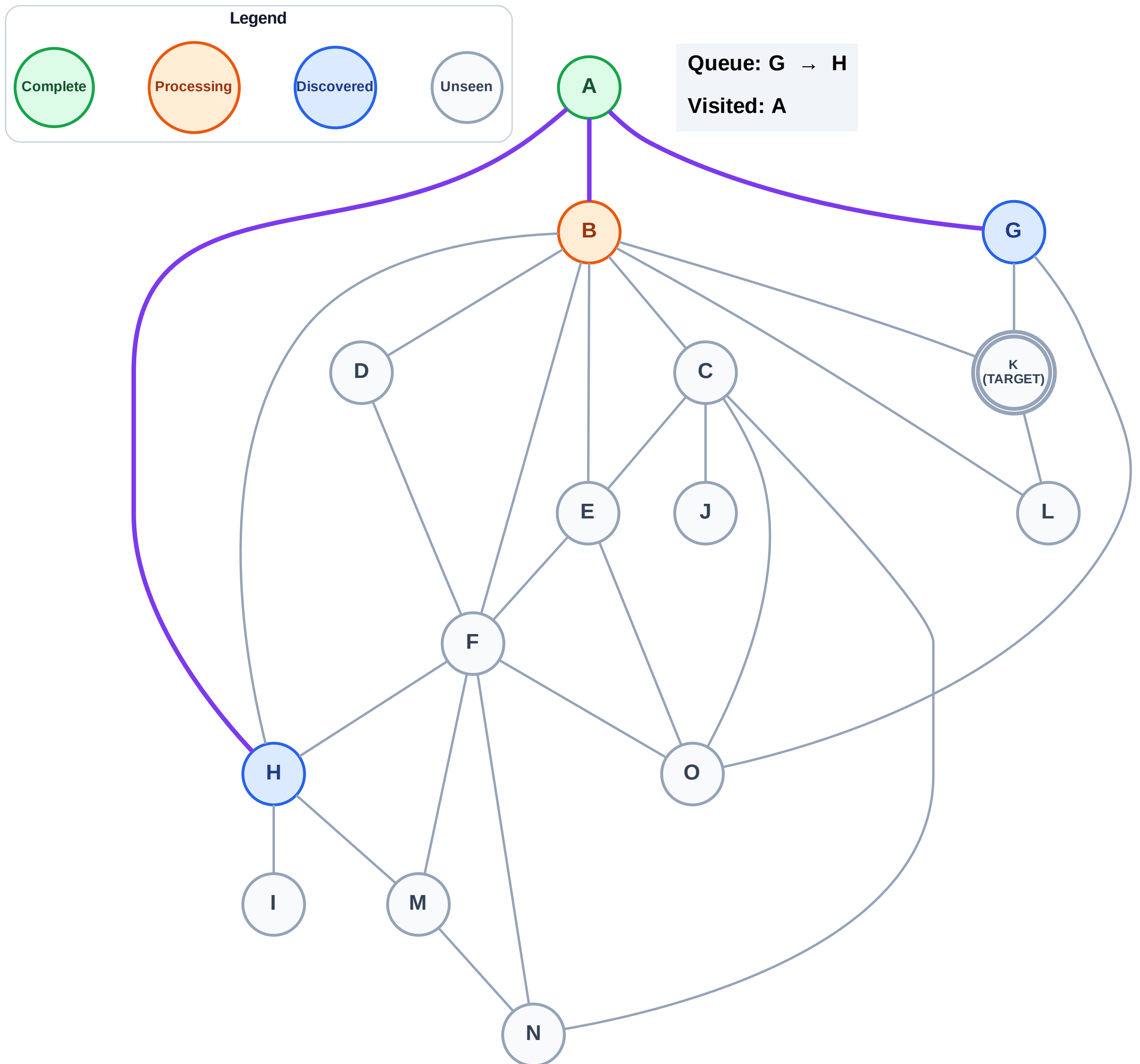
Queue: B → G → H

Visited: A



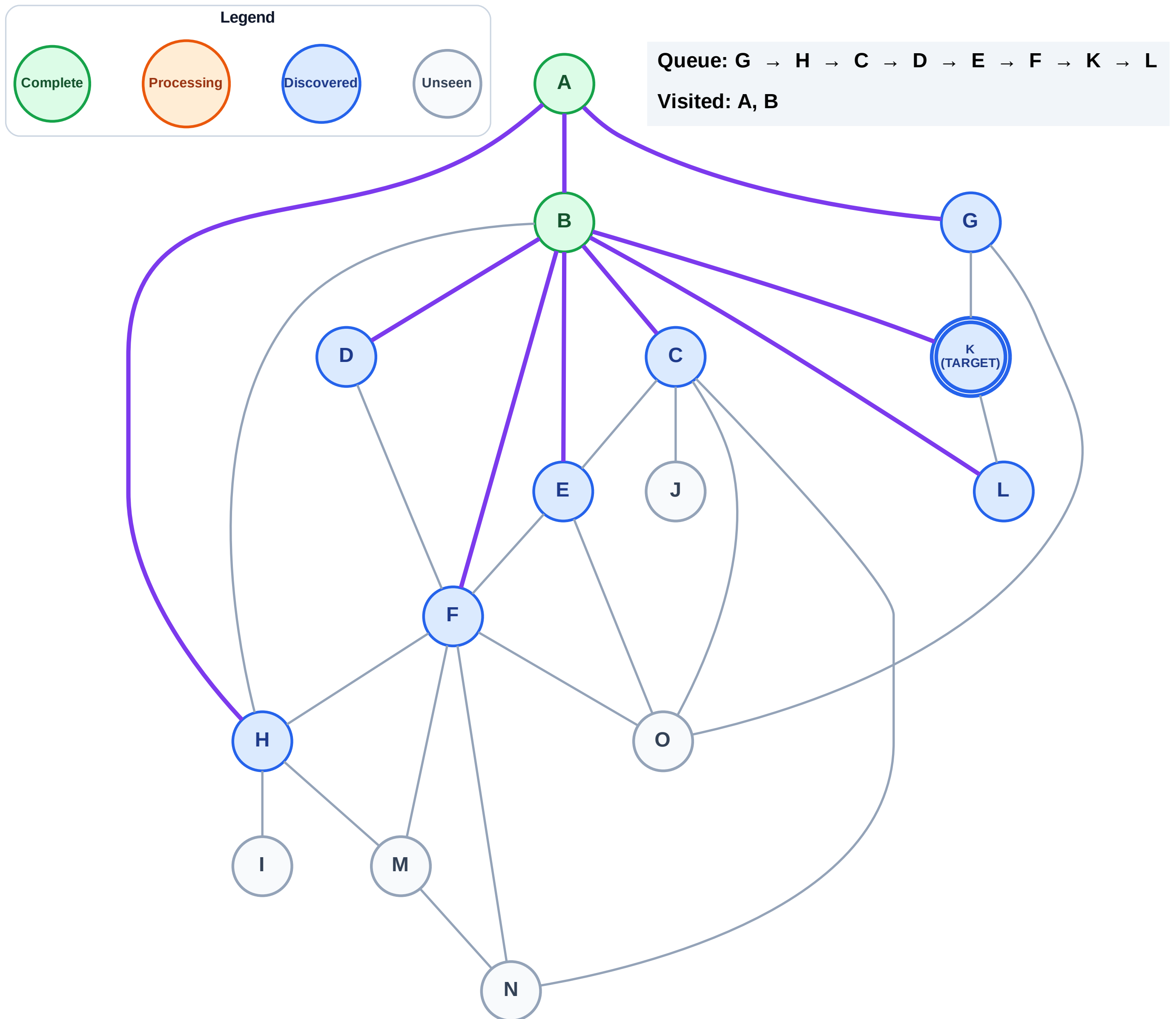
Step 4: Process node B

Step 4: Process node B



Step 5: Discover C, D, E, F, K, L from B

Step 5: Discover C, D, E, F, K, L from B



BFS Traversal

Step 6: Process node G

Legend

Complete

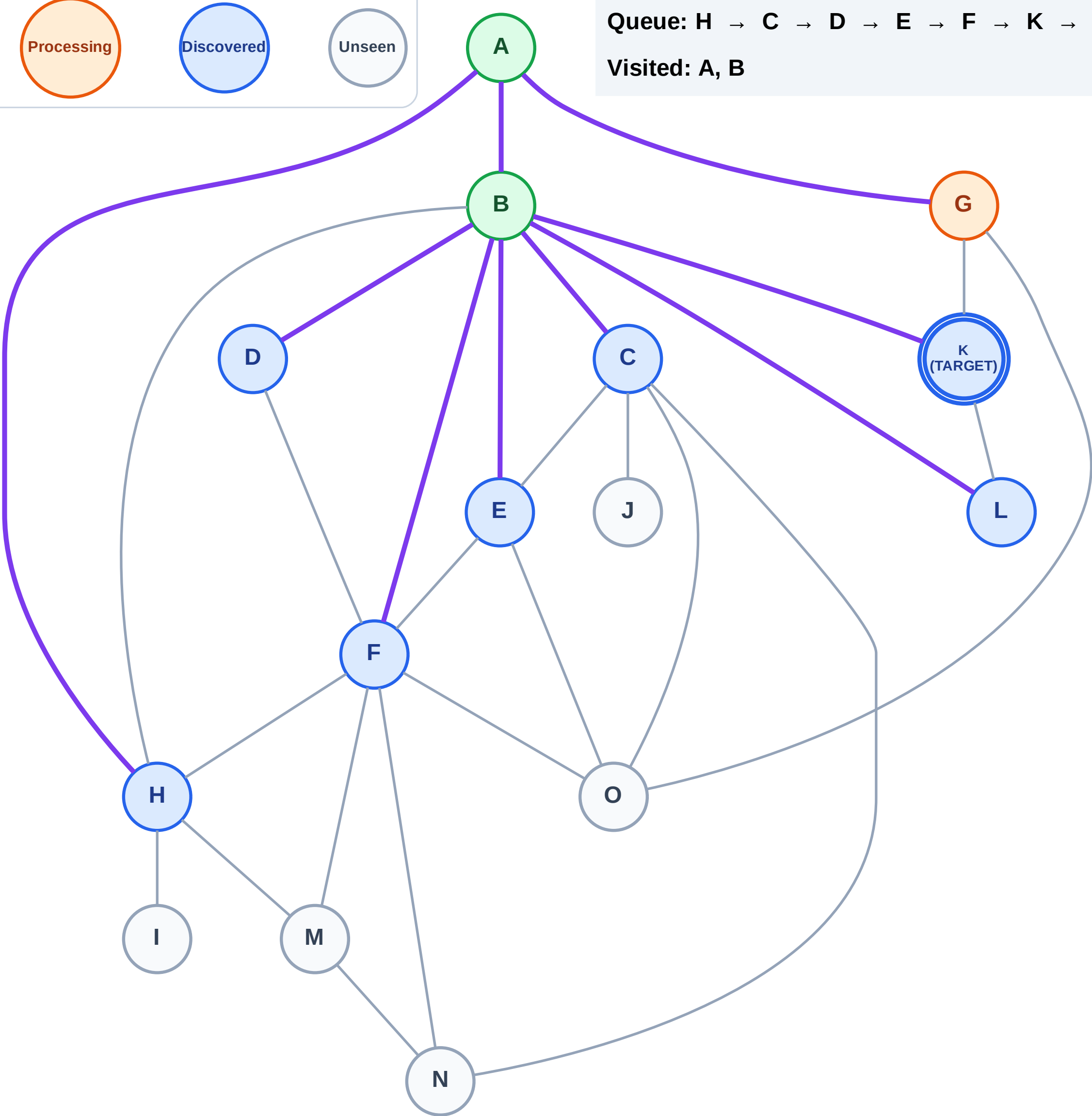
Processing

Discovered

Unseen

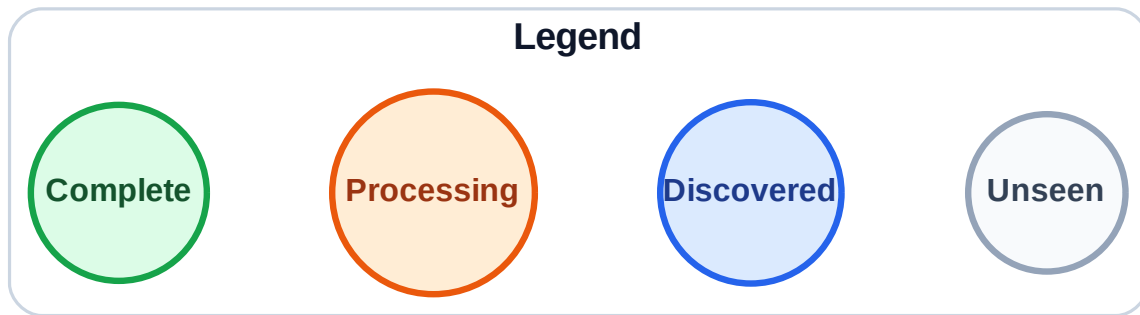
Queue: H → C → D → E → F → K → L

Visited: A, B



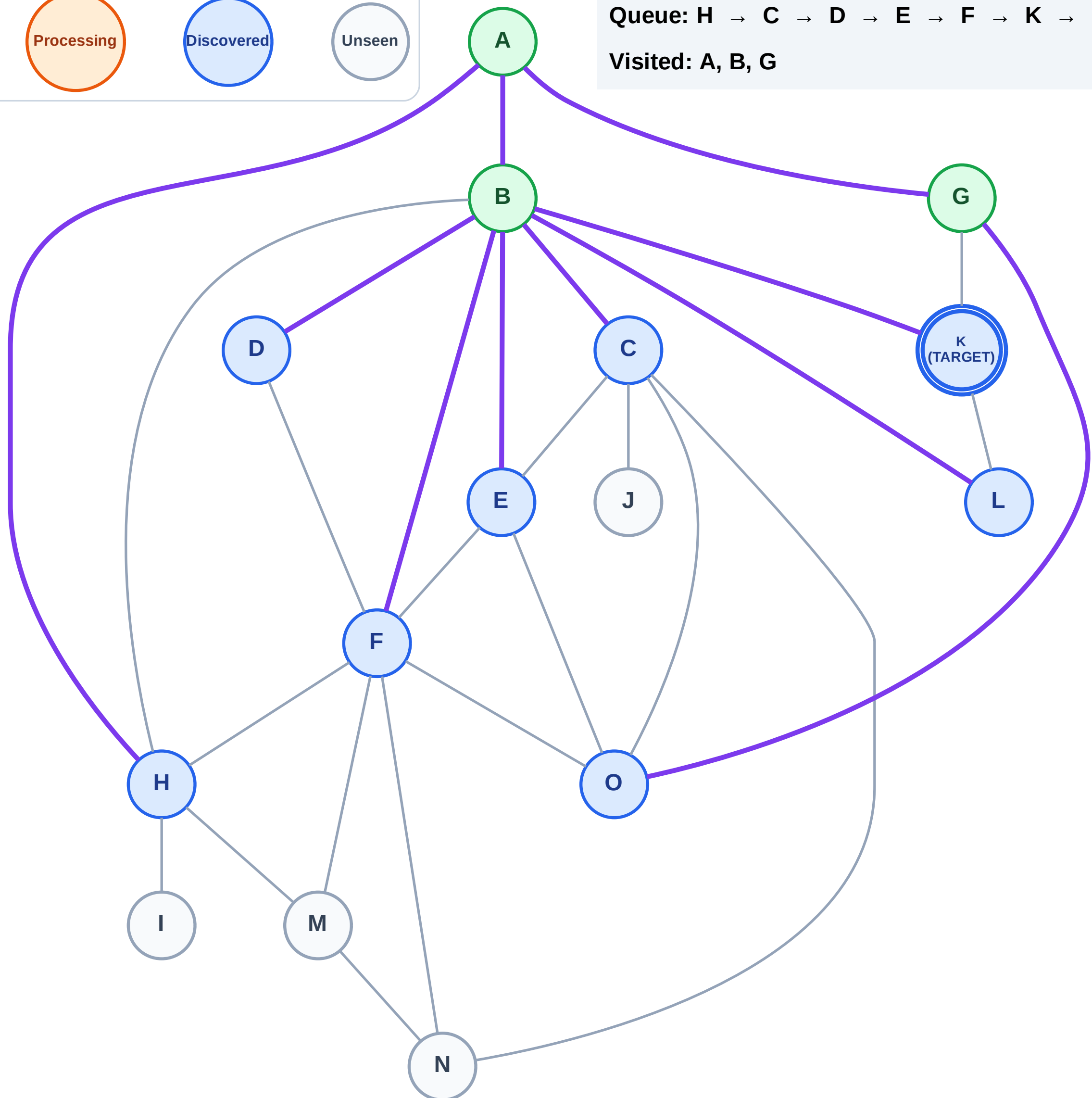
BFS Traversal

Step 7: Discover O from G



Queue: H → C → D → E → F → K → L → O

Visited: A, B, G



BFS Traversal

Step 8: Process node H

Legend

Complete

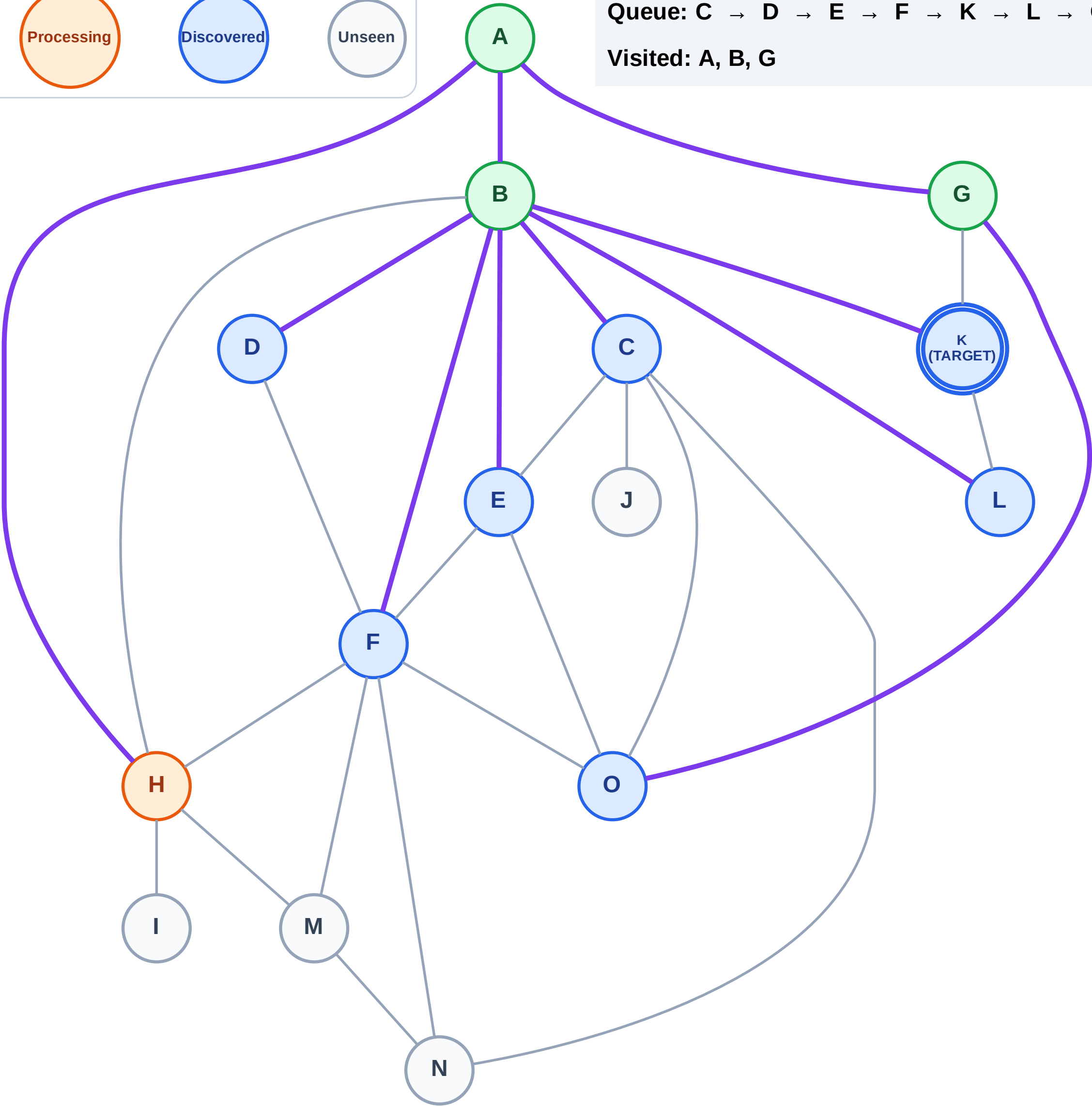
Processing

Discovered

Unseen

Queue: C → D → E → F → K → L → O

Visited: A, B, G



BFS Traversal

Step 9: Discover I, M from H

Legend



Complete



Processing



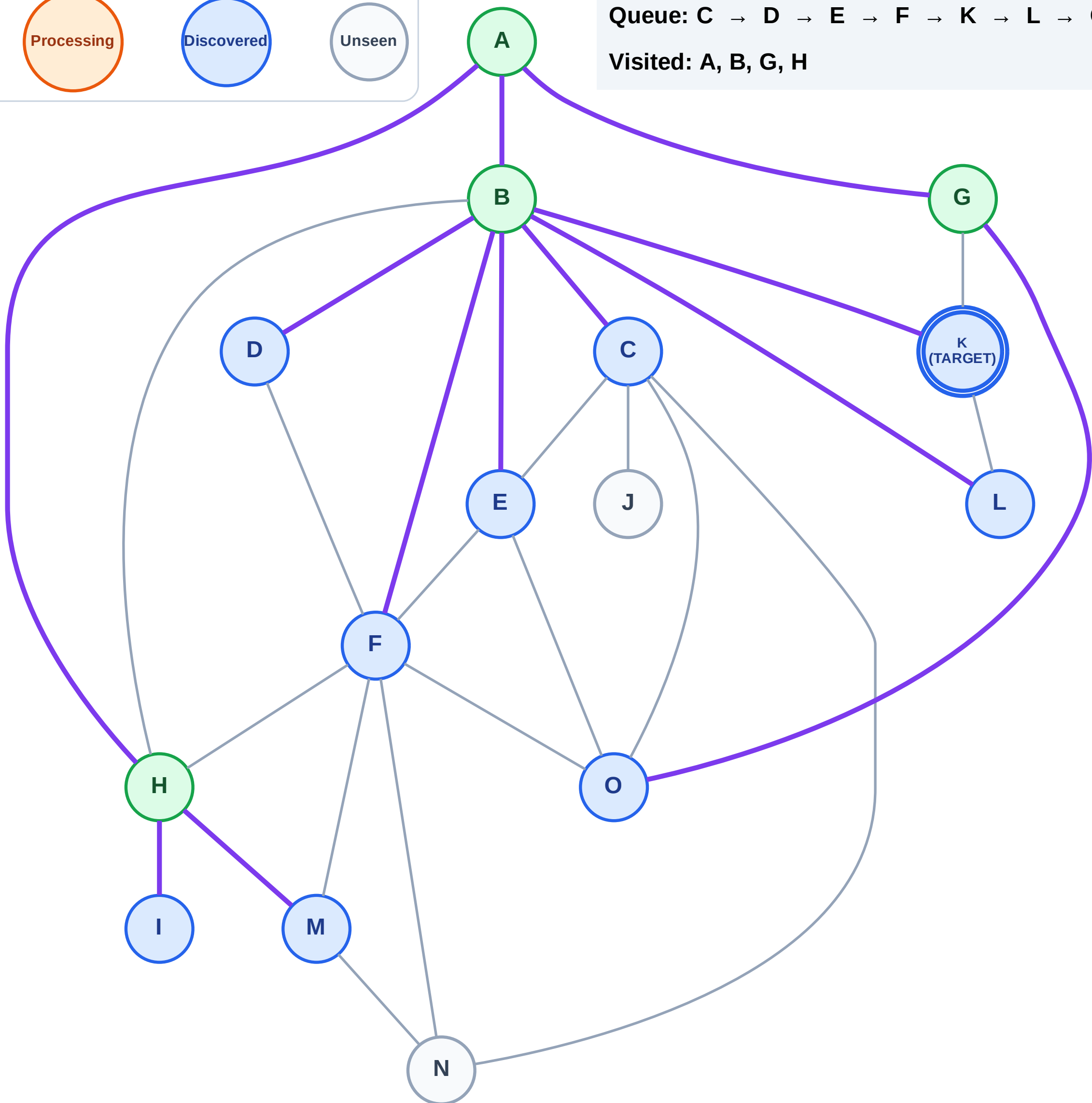
Discovered



Unseen

Queue: C → D → E → F → K → L → O → I → M

Visited: A, B, G, H



BFS Traversal

Step 10: Process node C

Legend

Complete

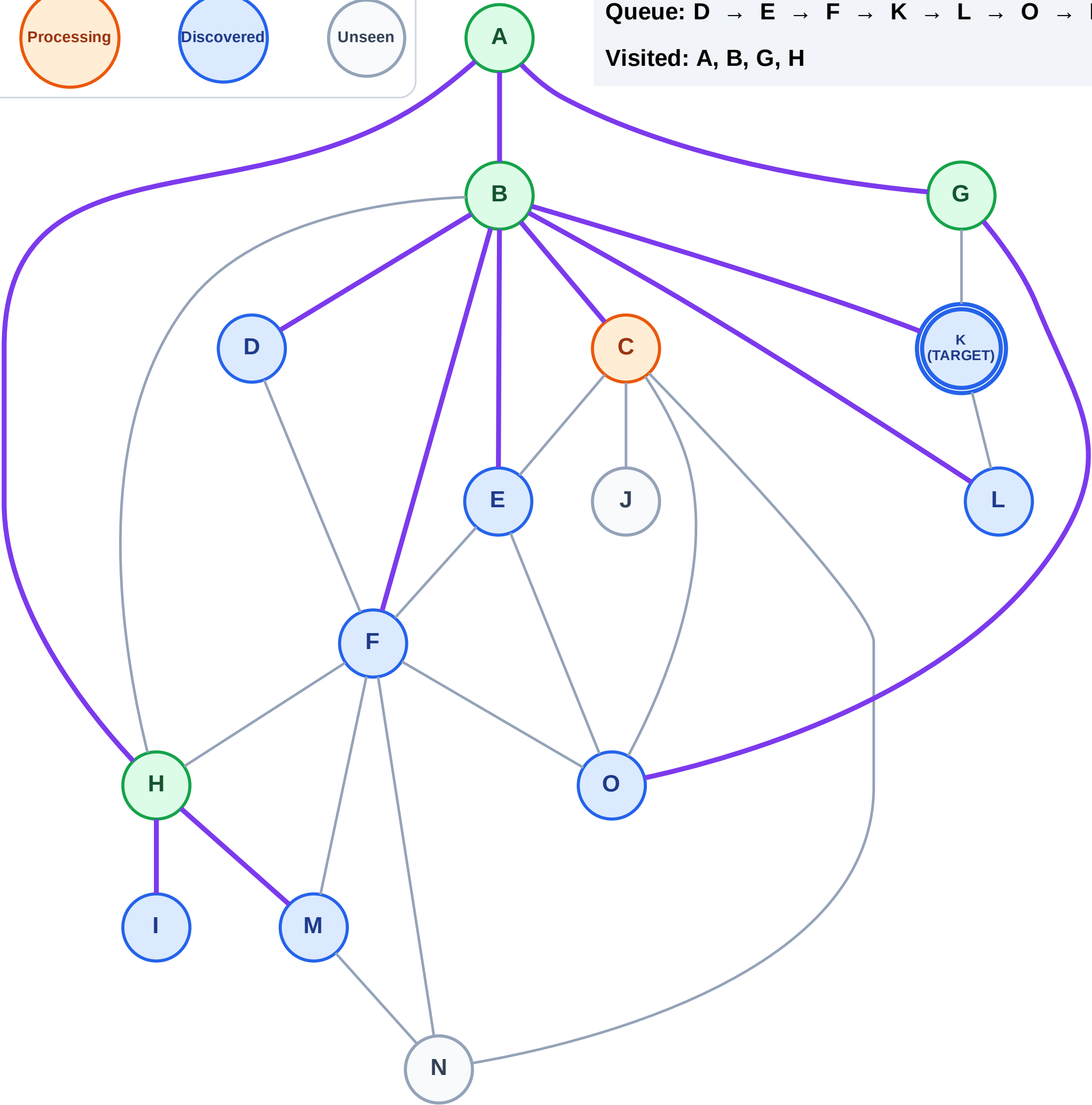
Processing

Discovered

Unseen

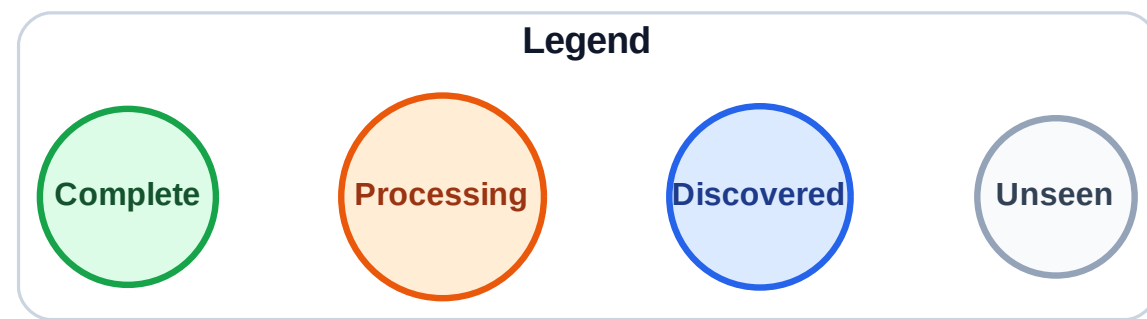
Queue: D → E → F → K → L → O → I → M

Visited: A, B, G, H



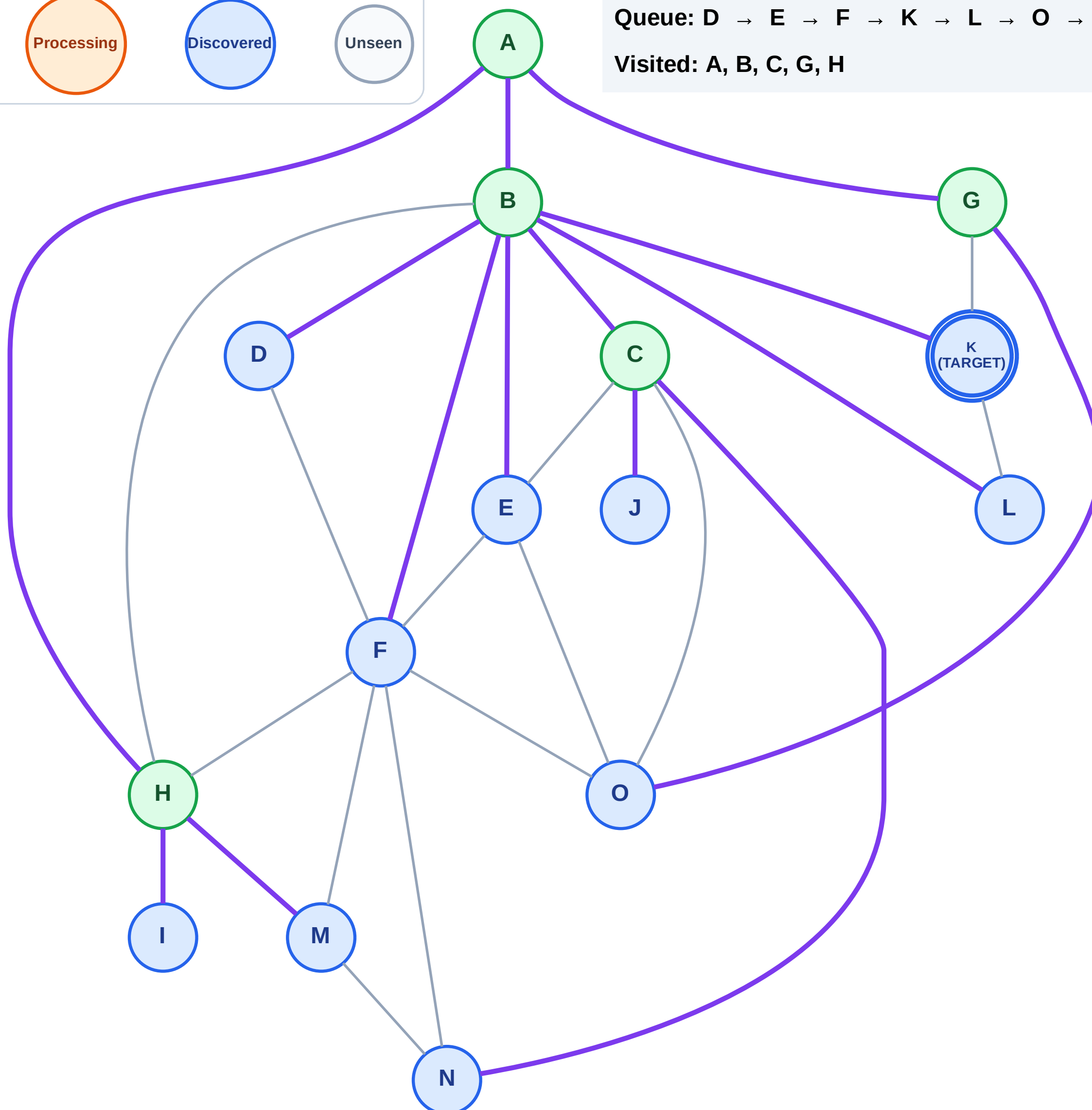
BFS Traversal

Step 11: Discover J, N from C



Queue: D → E → F → K → L → O → I → M → J → N

Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node D

Legend

Complete

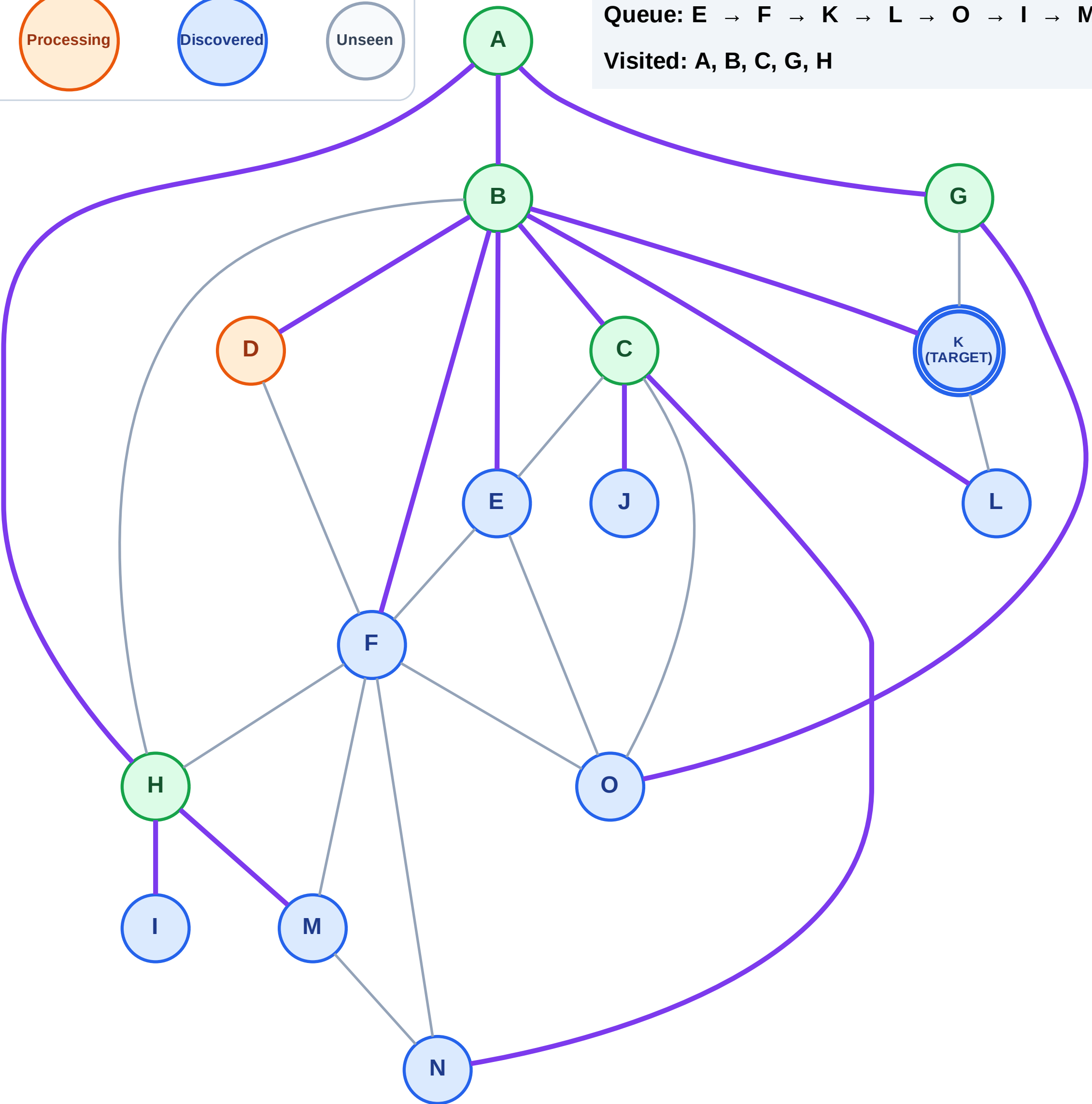
Processing

Discovered

Unseen

Queue: E → F → K → L → O → I → M → J → N

Visited: A, B, C, G, H



BFS Traversal

Step 13: No new neighbors from D

Legend

Complete

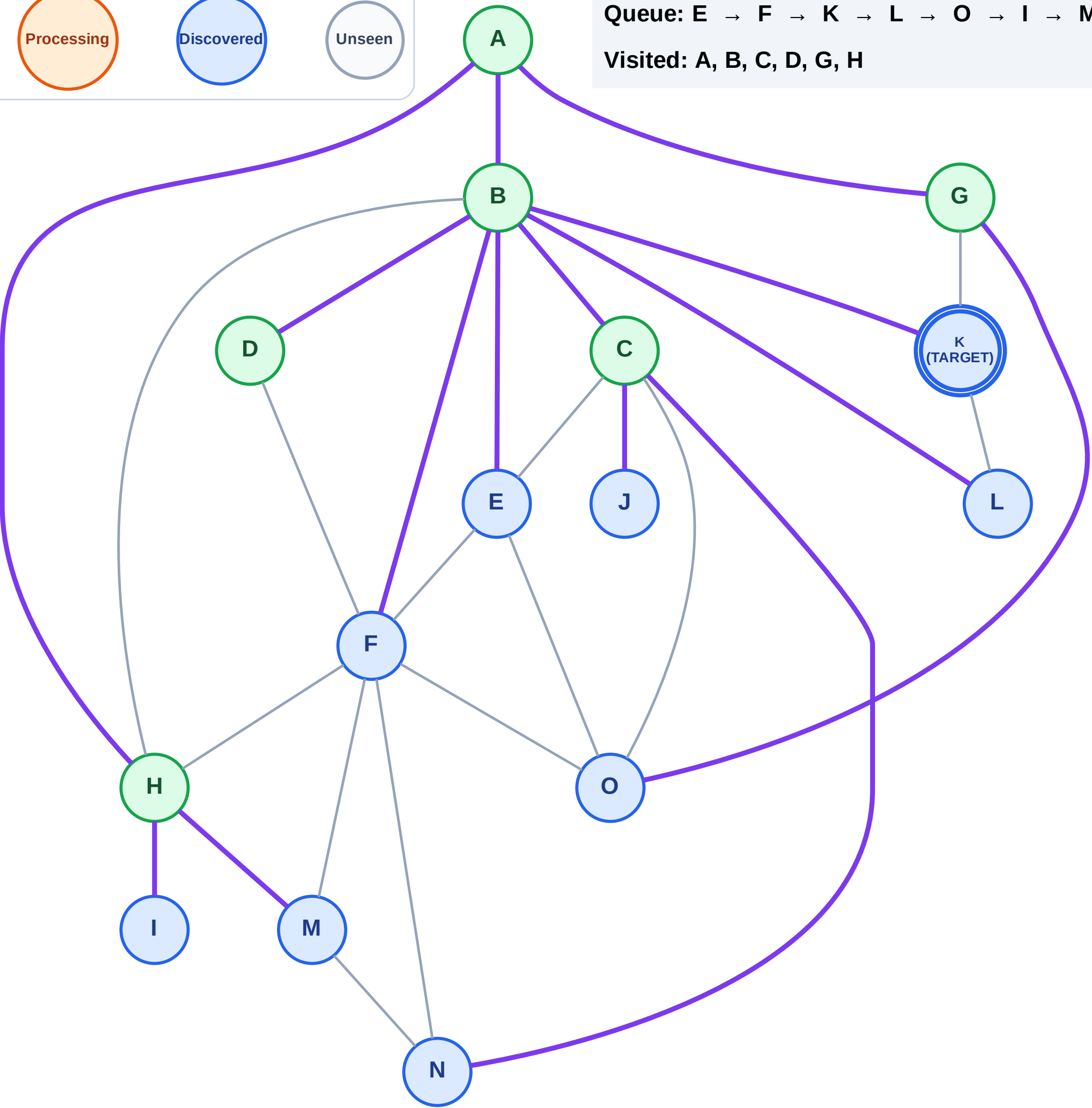
Processing

Discovered

Unseen

Queue: E → F → K → L → O → I → M → J → N

Visited: A, B, C, D, G, H



BFS Traversal

Step 14: Process node E

Legend

Complete

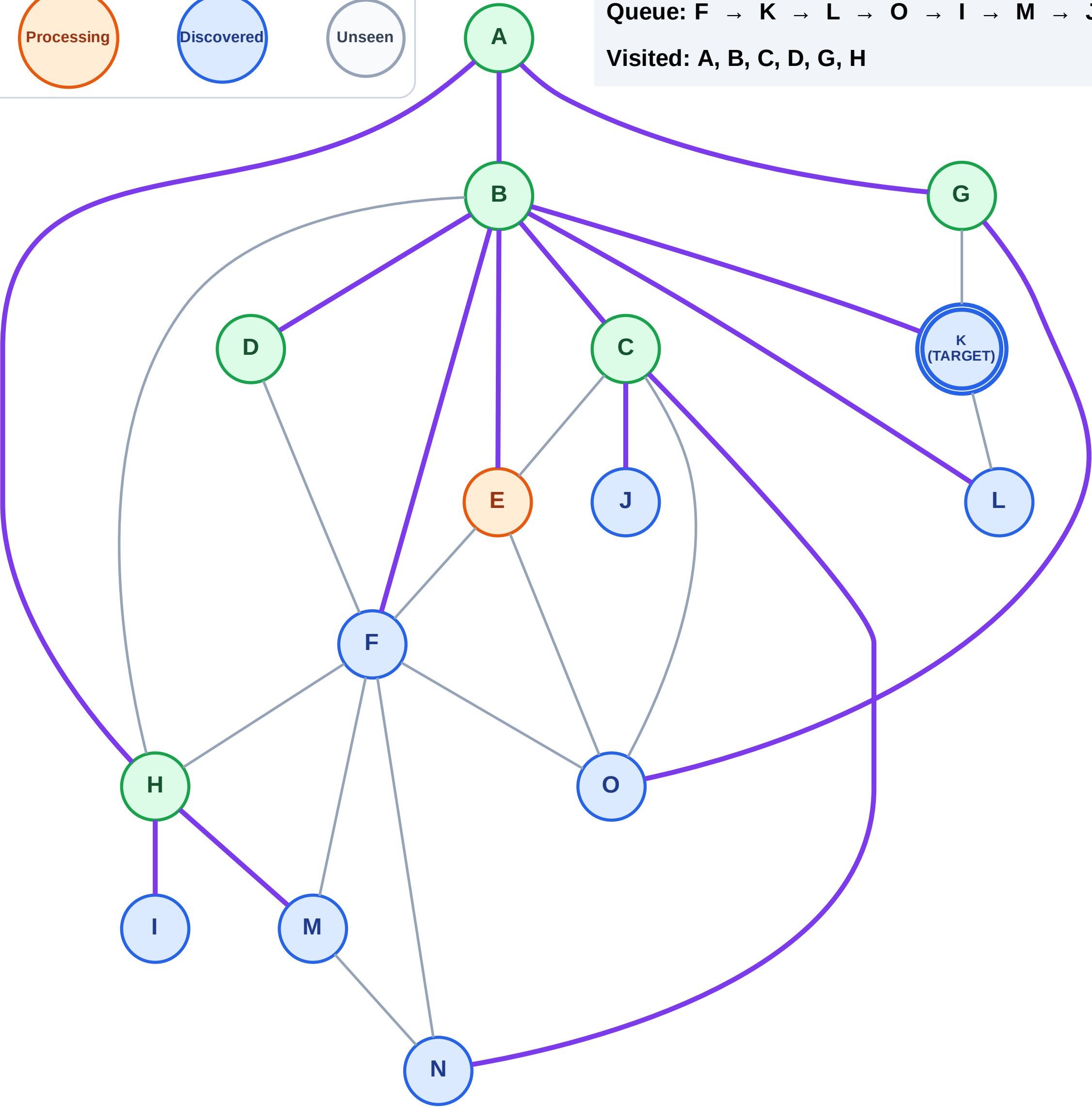
Processing

Discovered

Unseen

Queue: F → K → L → O → I → M → J → N

Visited: A, B, C, D, G, H



BFS Traversal

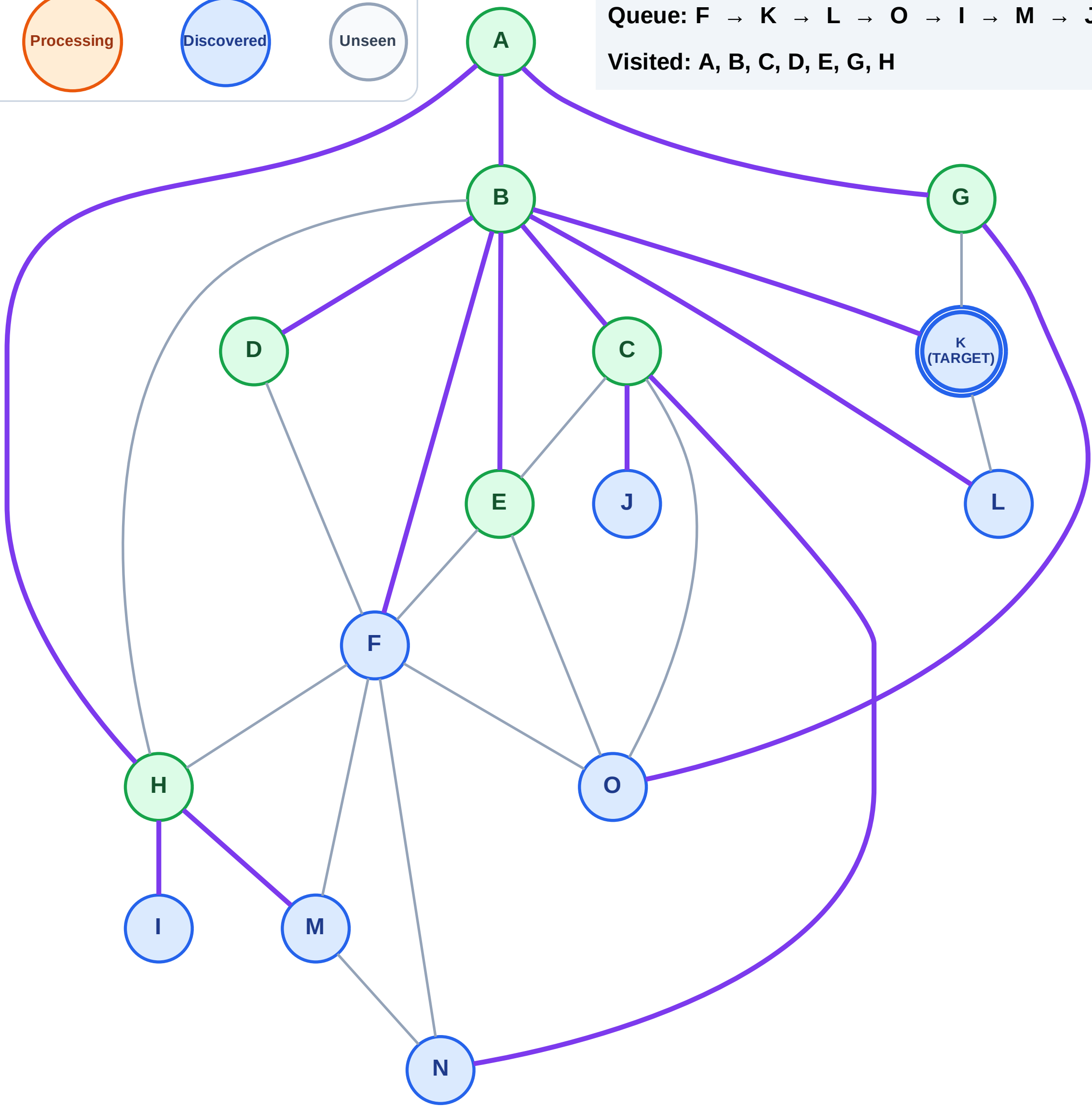
Step 15: No new neighbors from E

Legend



Queue: F → K → L → O → I → M → J → N

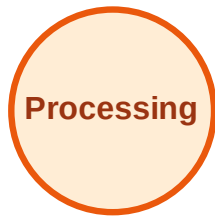
Visited: A, B, C, D, E, G, H



BFS Traversal

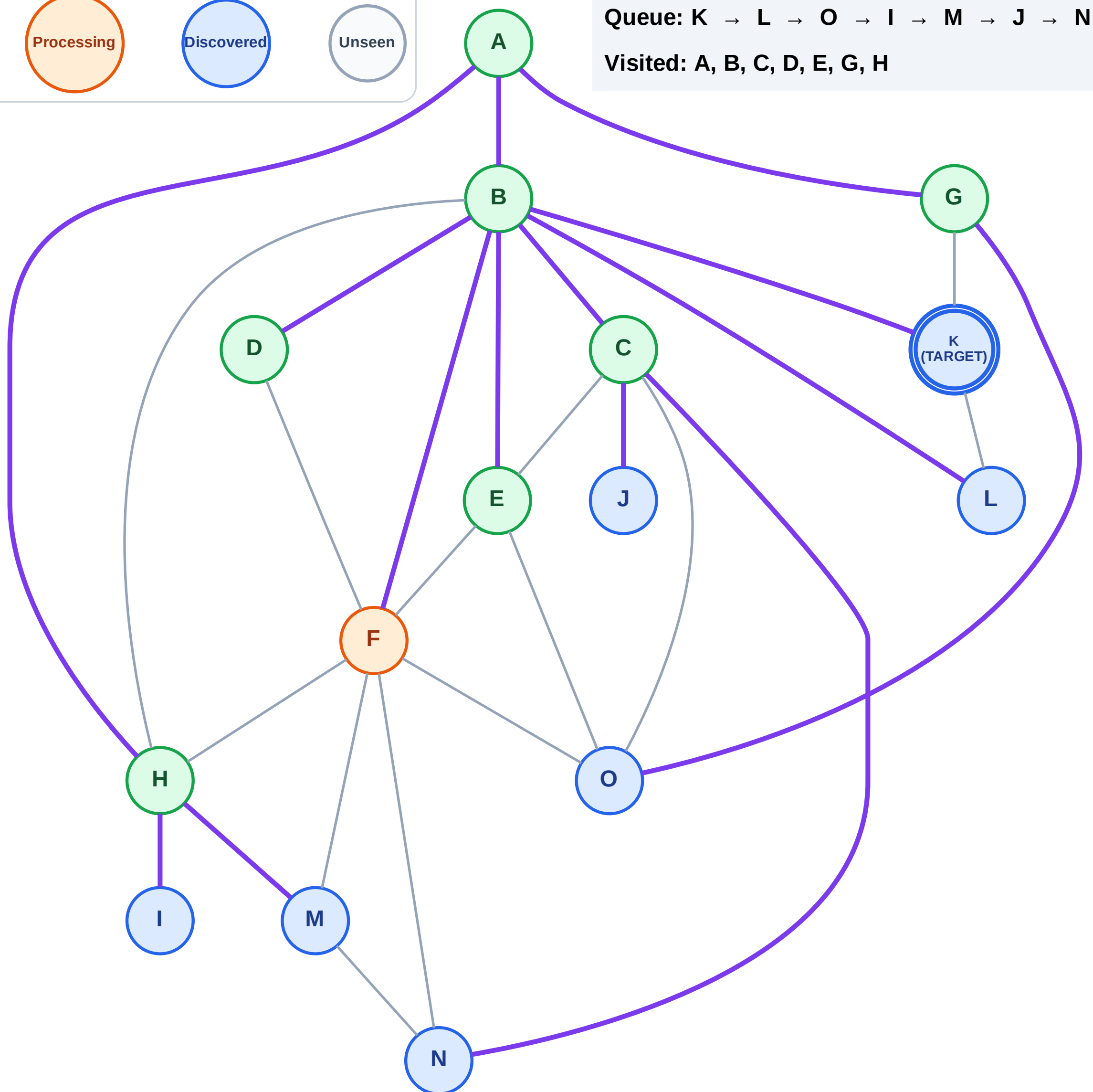
Step 16: Process node F

Legend



Queue: K → L → O → I → M → J → N

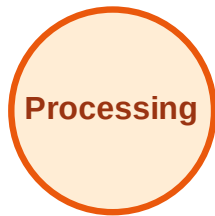
Visited: A, B, C, D, E, G, H



BFS Traversal

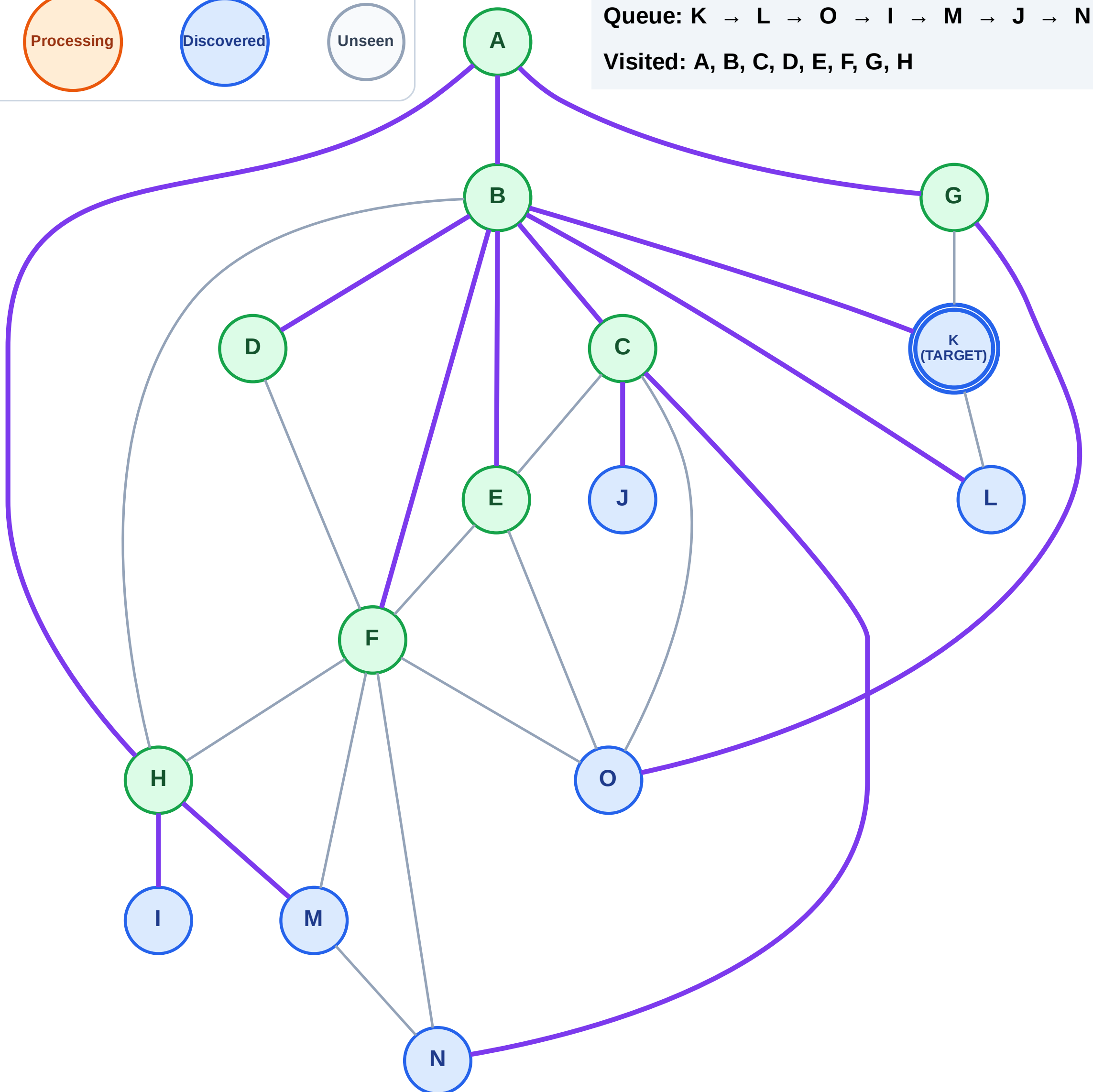
Step 17: No new neighbors from F

Legend



Queue: K → L → O → I → M → J → N

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node K

Legend

Complete

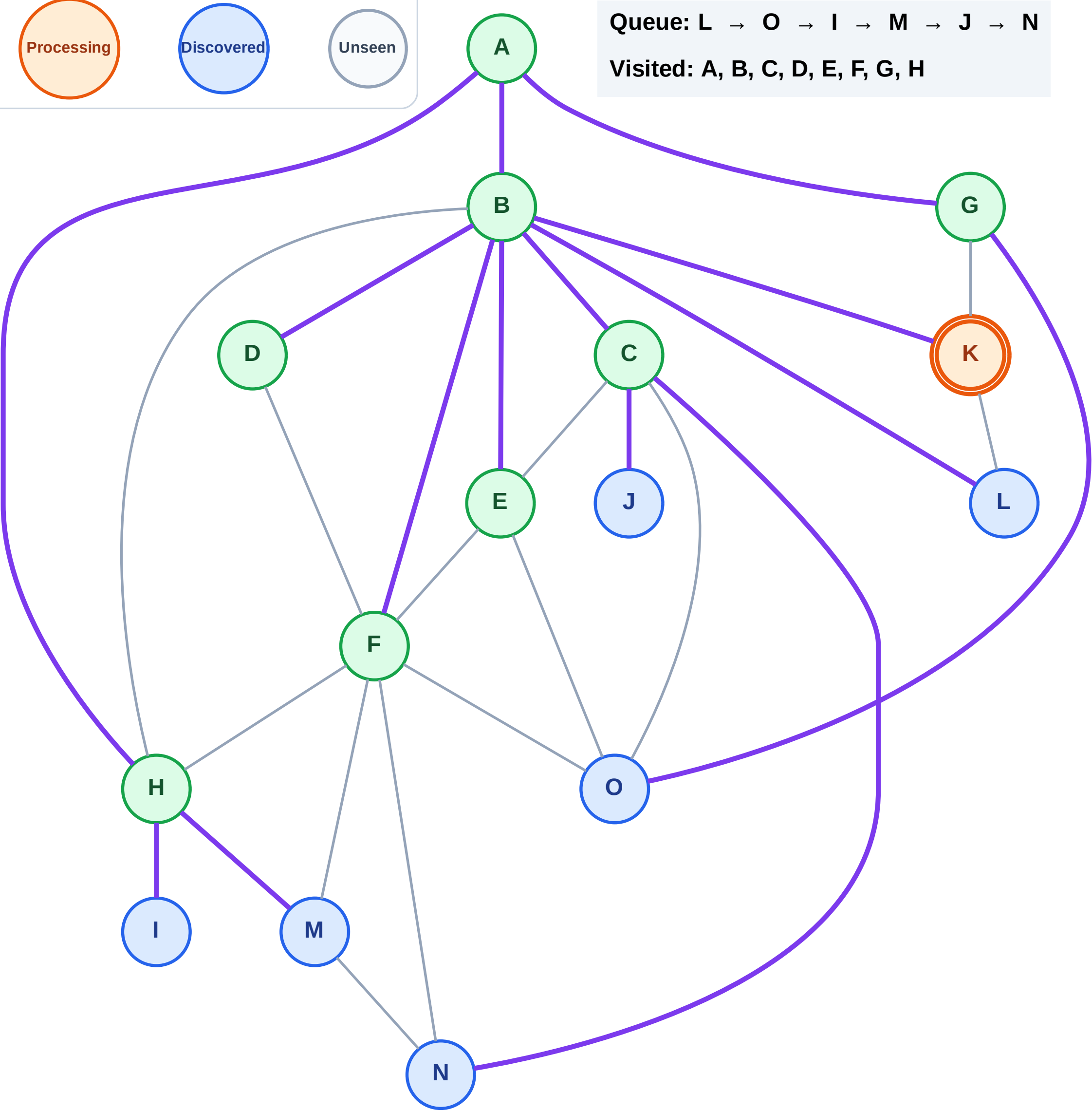
Processing

Discovered

Unseen

Queue: L → O → I → M → J → N

Visited: A, B, C, D, E, F, G, H



BFS Traversal

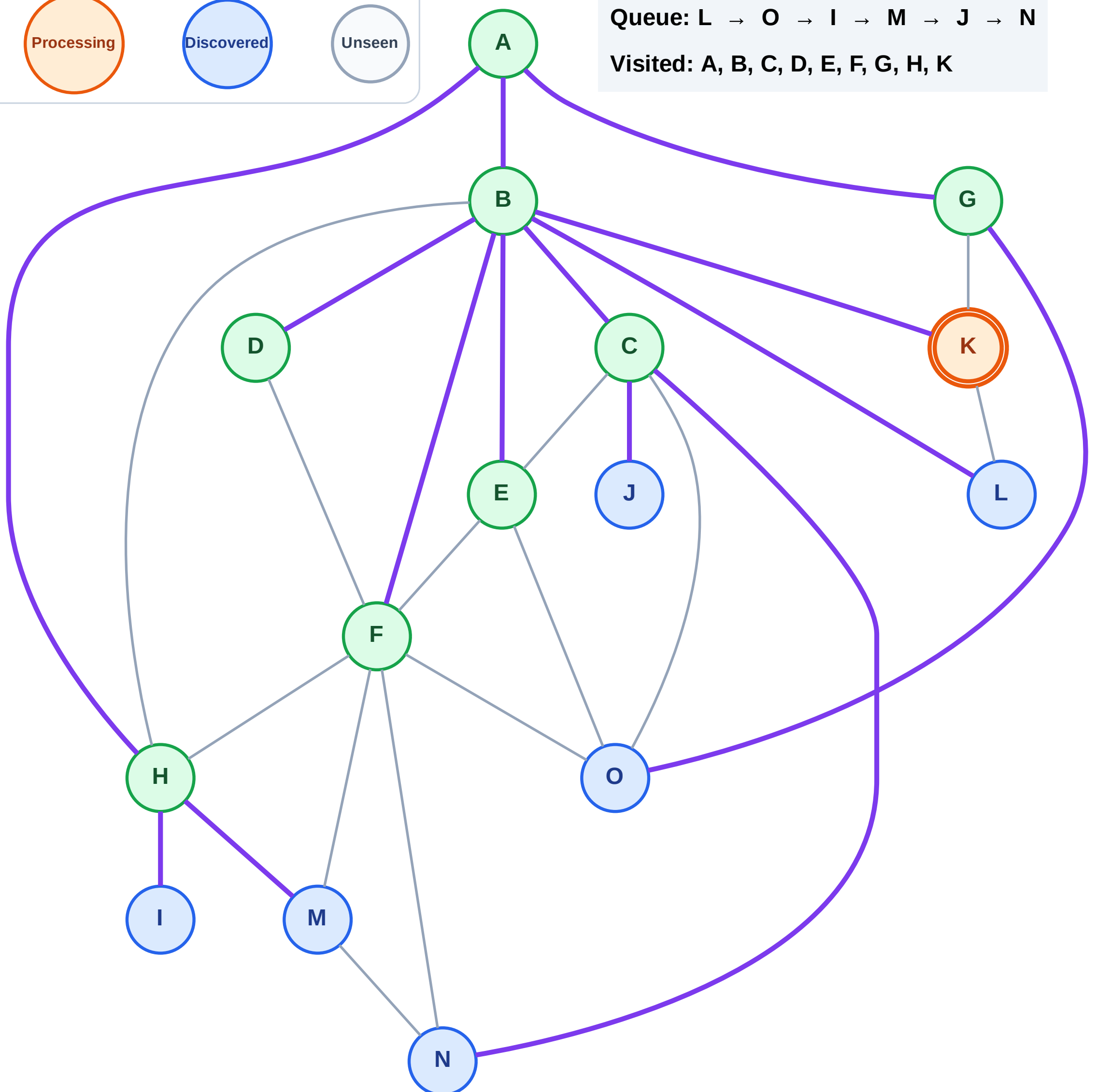
Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Queue: L → O → I → M → J → N

Visited: A, B, C, D, E, F, G, H, K



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

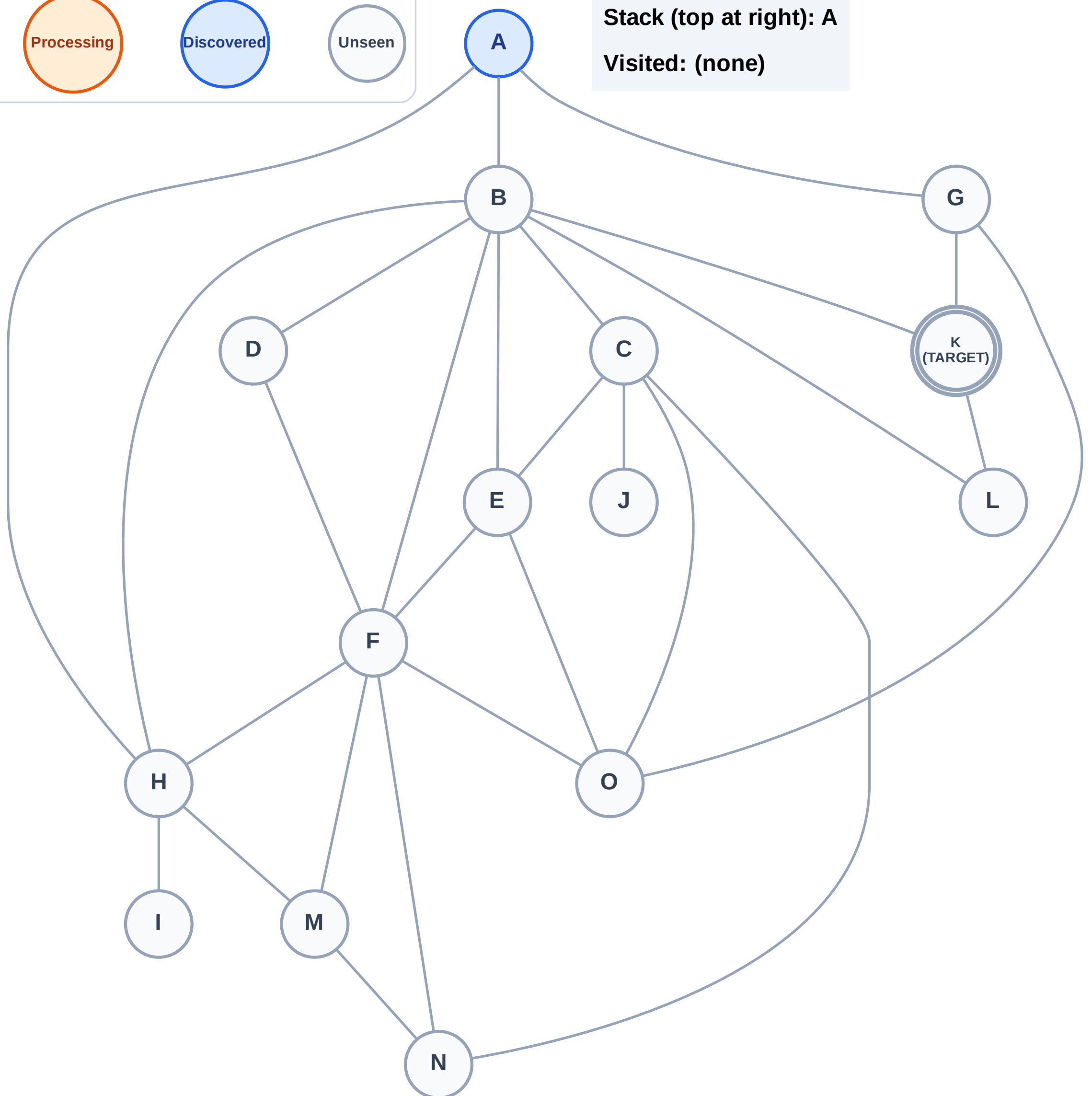
Processing

Discovered

Unseen

Stack (top at right): A

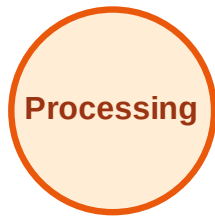
Visited: (none)



DFS Traversal

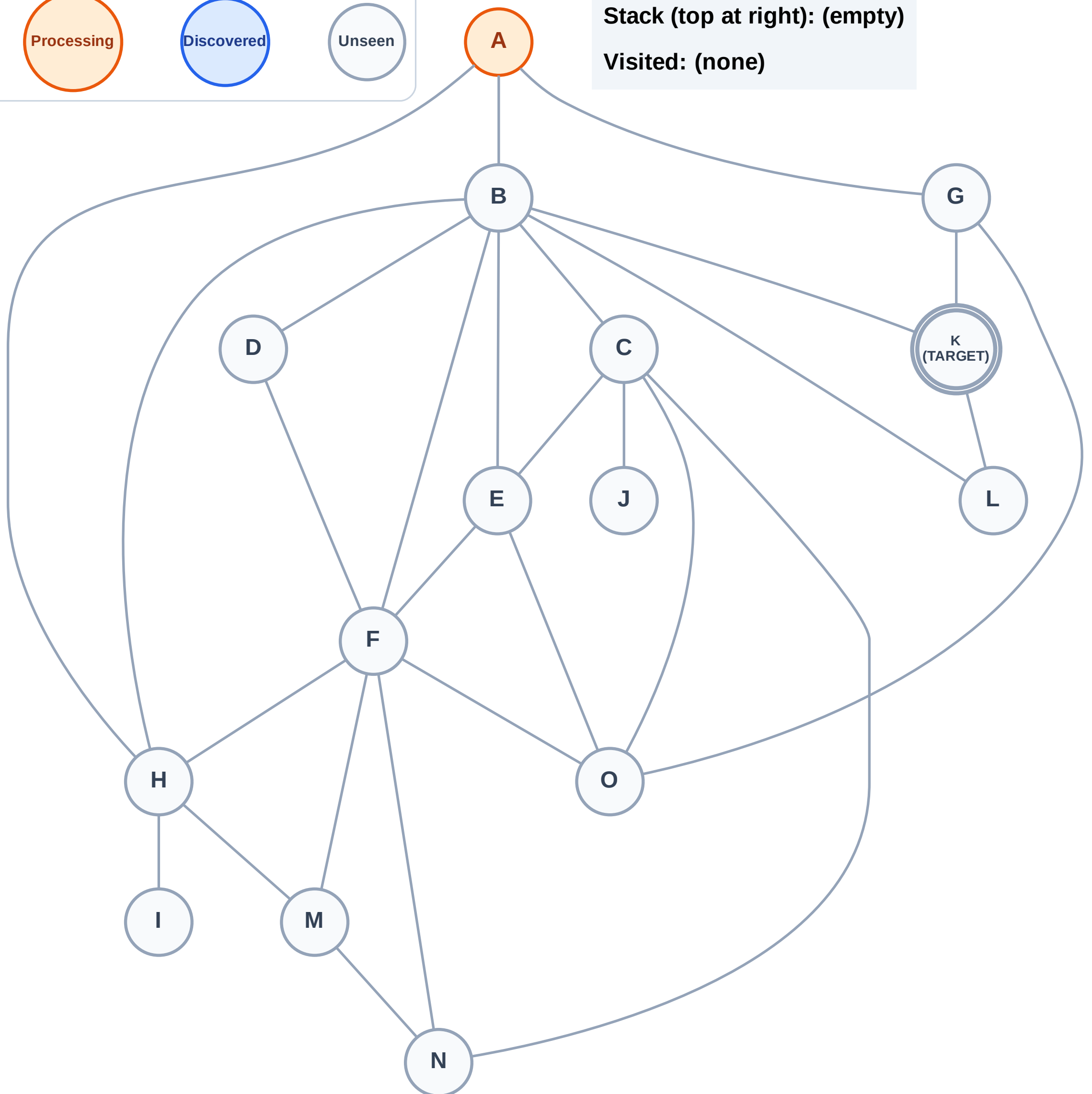
Step 2: Process node A

Legend



Stack (top at right): (empty)

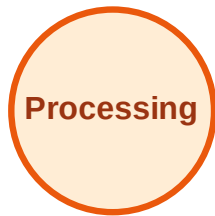
Visited: (none)



DFS Traversal

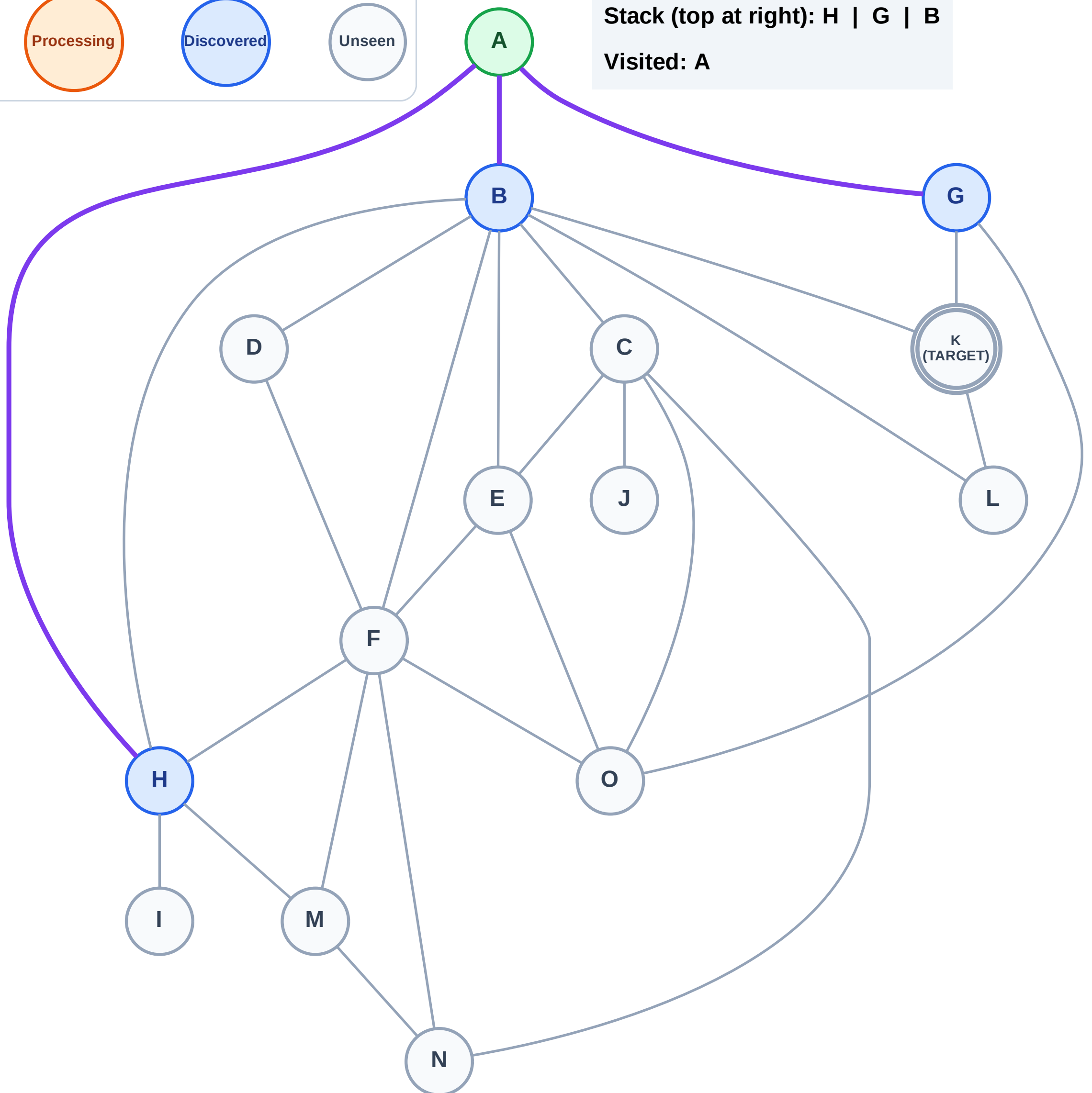
Step 3: Discover H, G, B from A

Legend



Stack (top at right): H | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

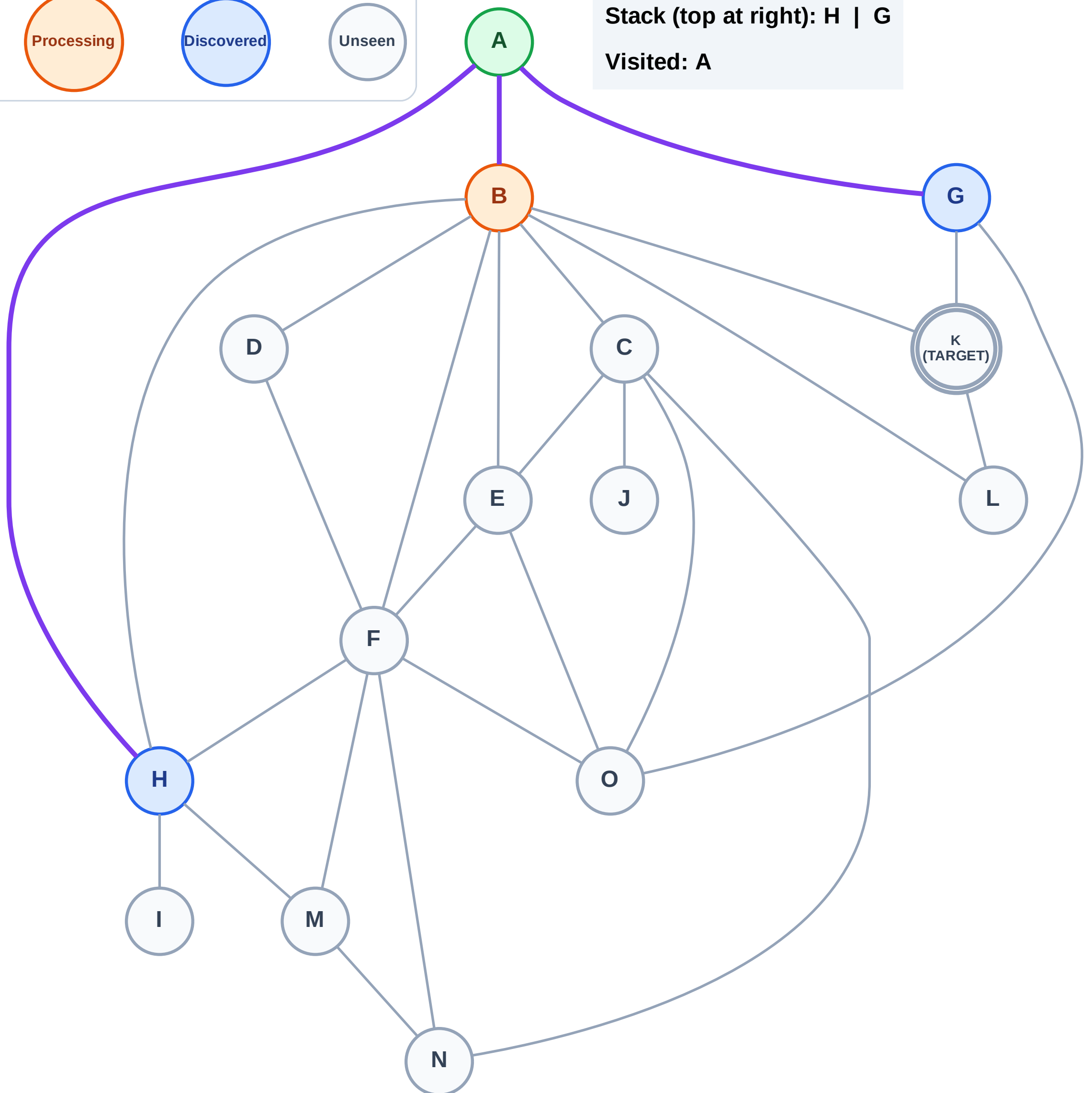
Processing

Discovered

Unseen

Stack (top at right): H | G

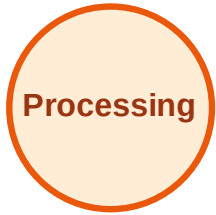
Visited: A



DFS Traversal

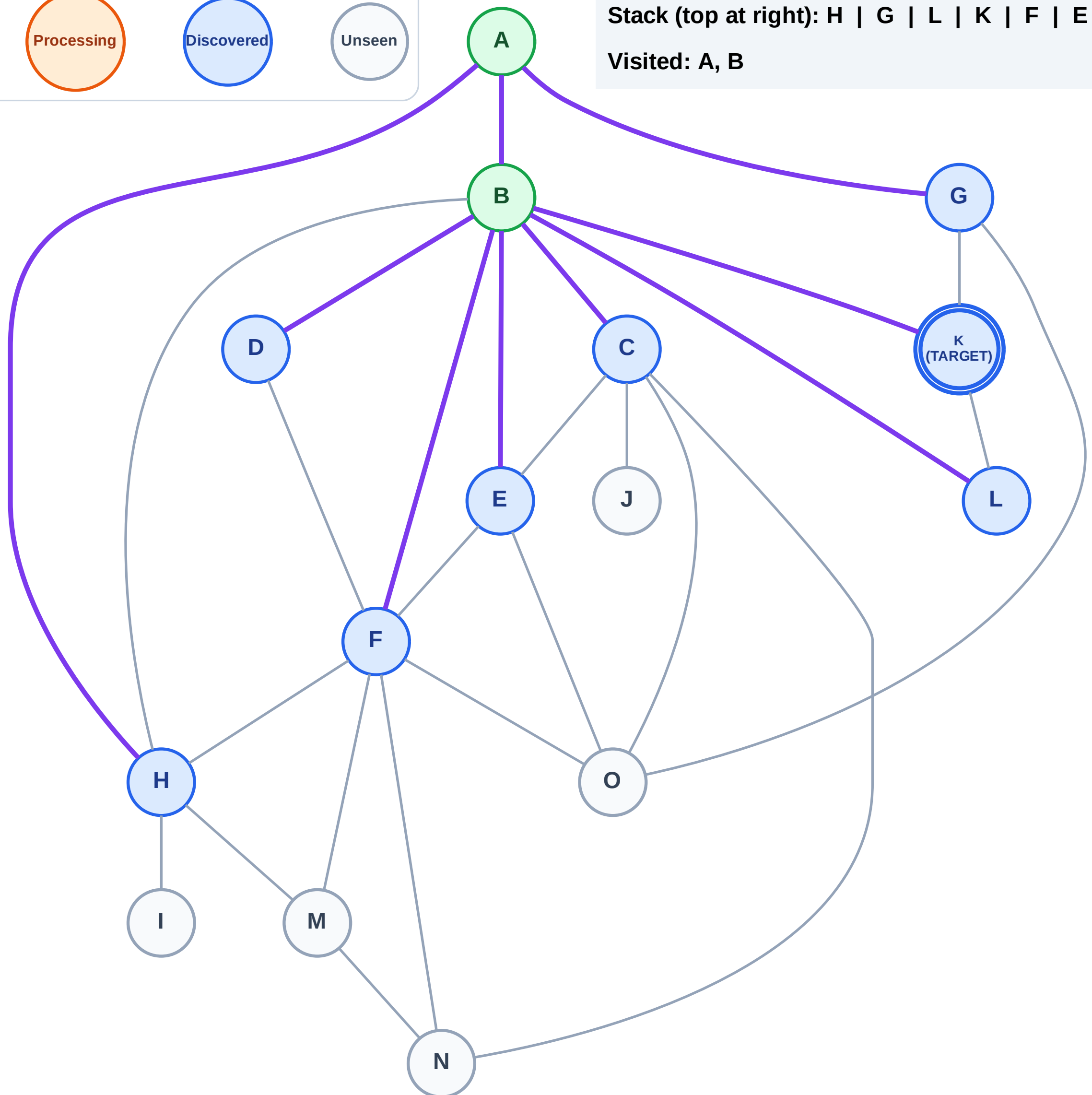
Step 5: Discover L, K, F, E, D, C from B

Legend



Stack (top at right): H | G | L | K | F | E | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

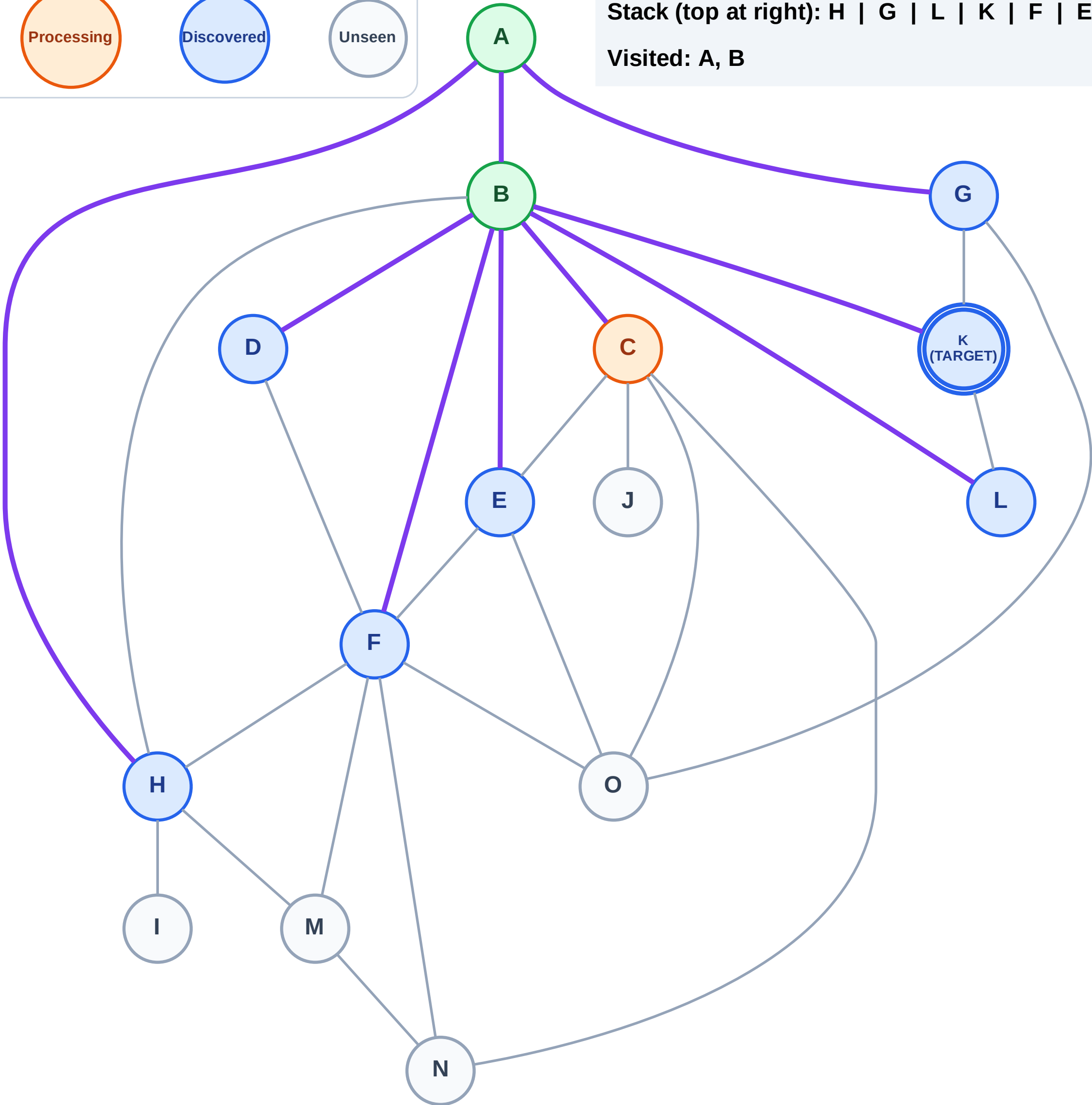
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K | F | E | D

Visited: A, B

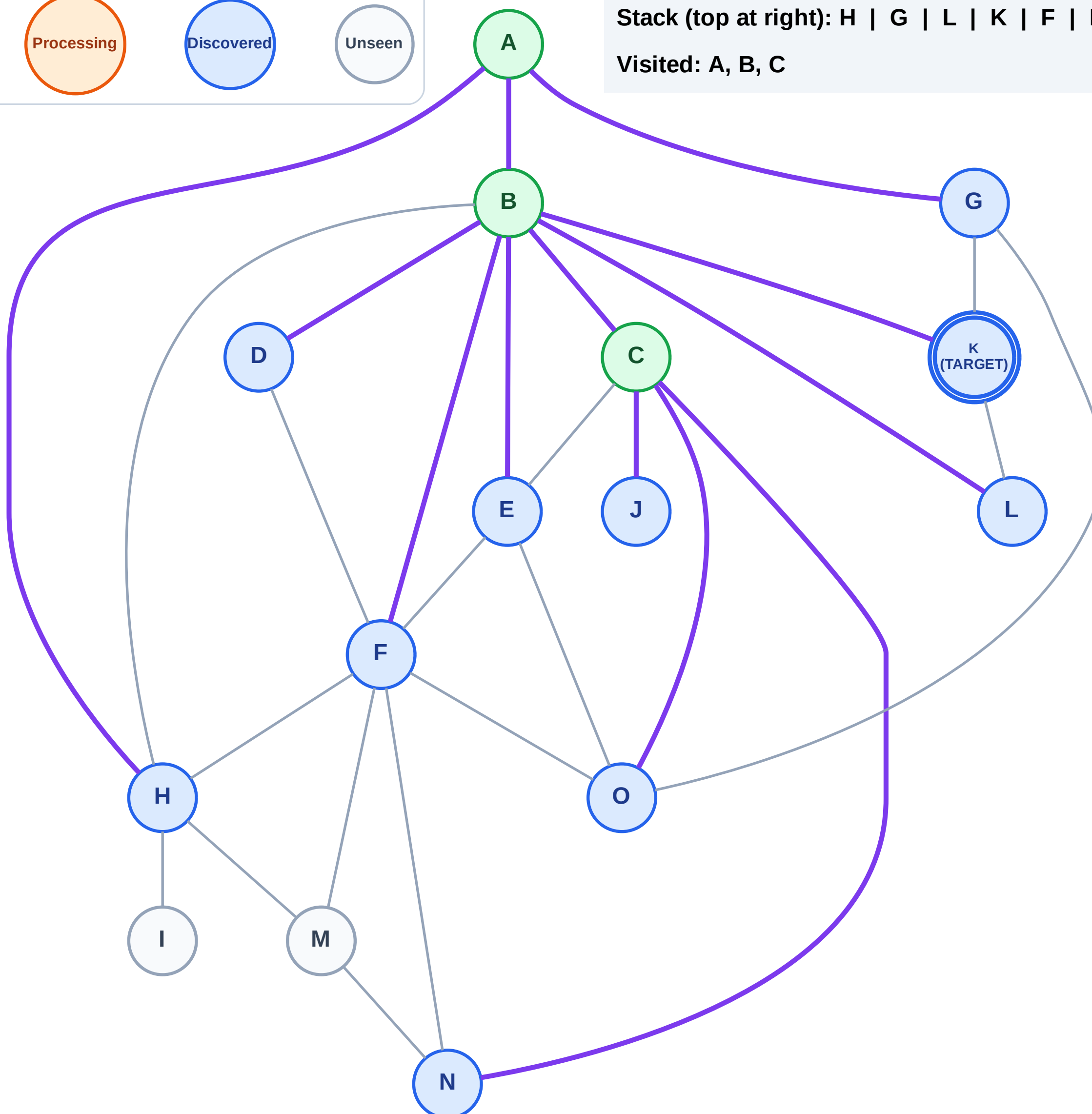


Step 7: Discover O, N, J from C

Stack (top at right): H | G | L | K | F | E | D | O | N | J

Legend

Unseen



DFS Traversal

Step 8: Process node J

Legend

Complete

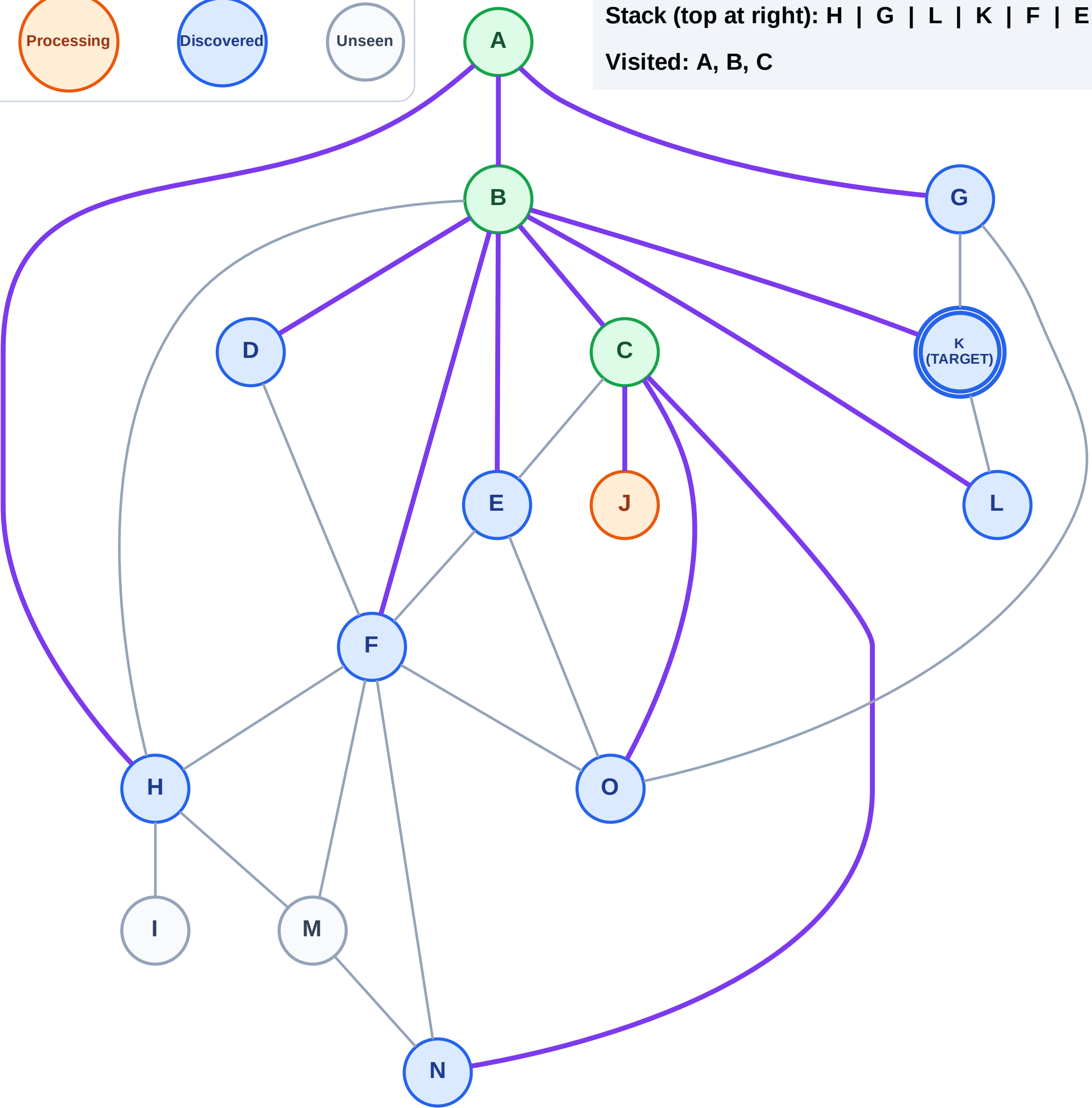
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K | F | E | D | O | N

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from J

Legend

Complete

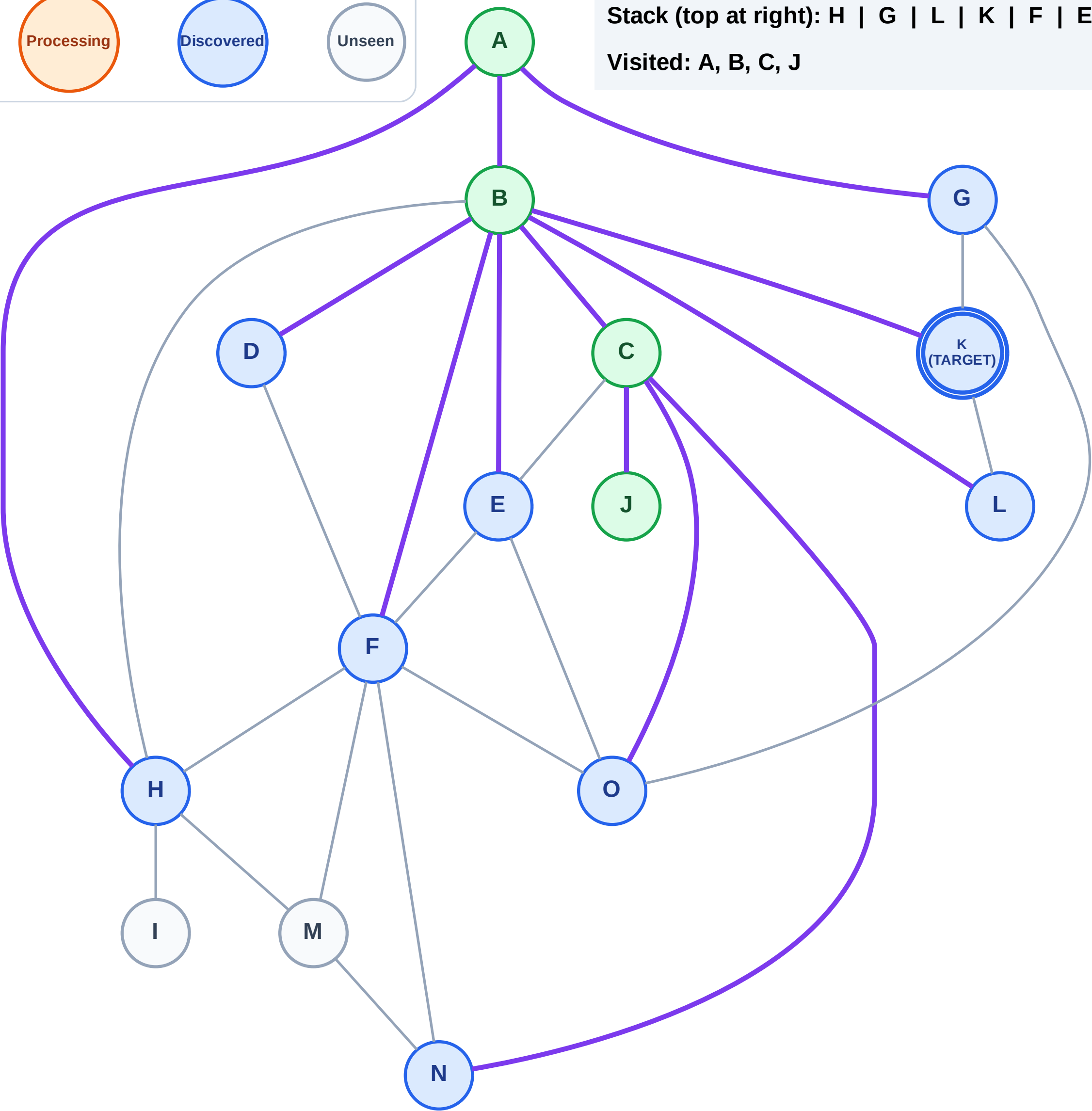
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K | F | E | D | O | N

Visited: A, B, C, J



DFS Traversal

Step 10: Process node N

Legend

Complete

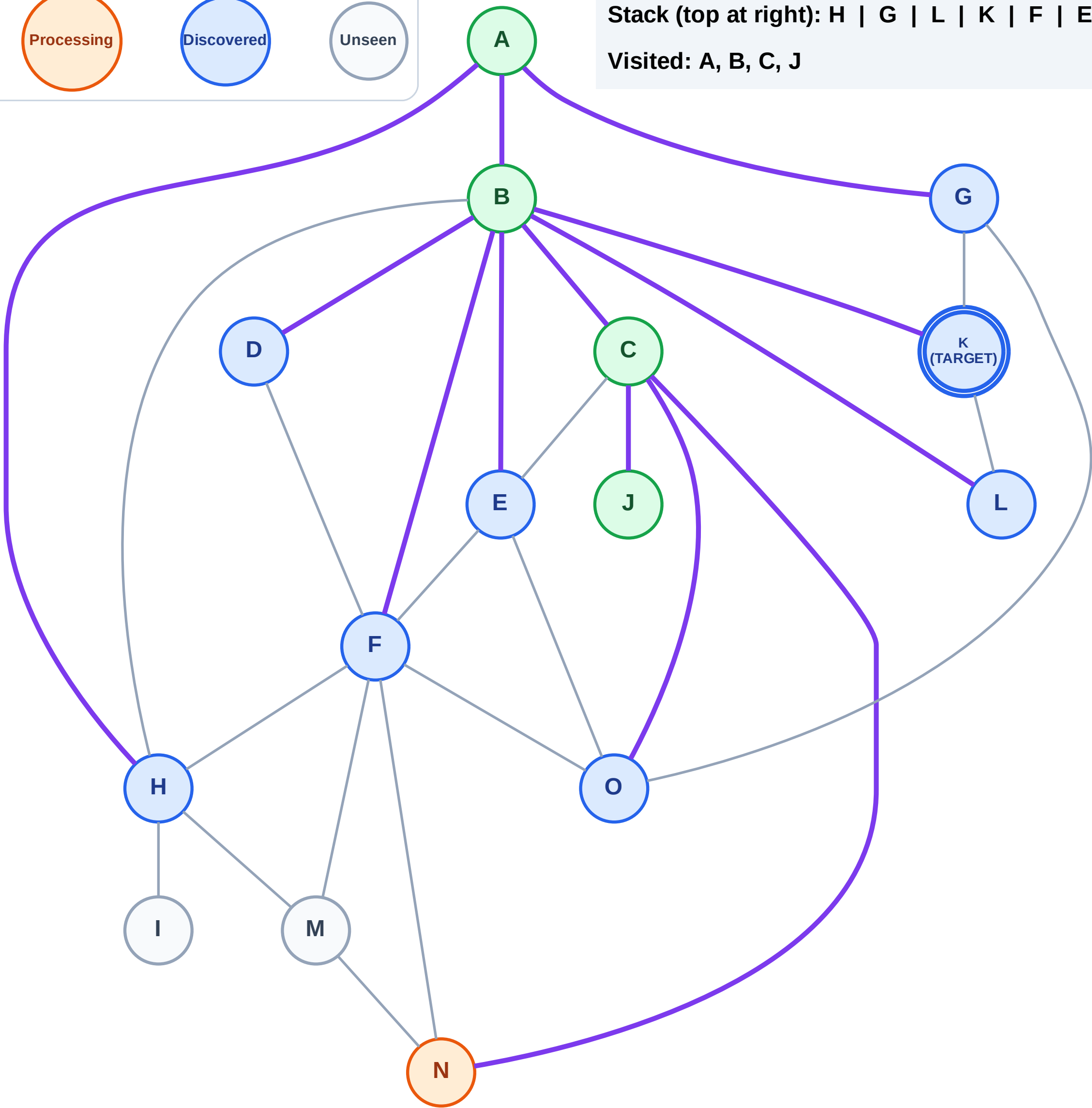
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K | F | E | D | O

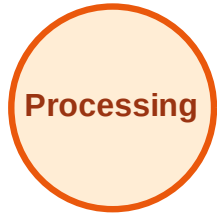
Visited: A, B, C, J



DFS Traversal

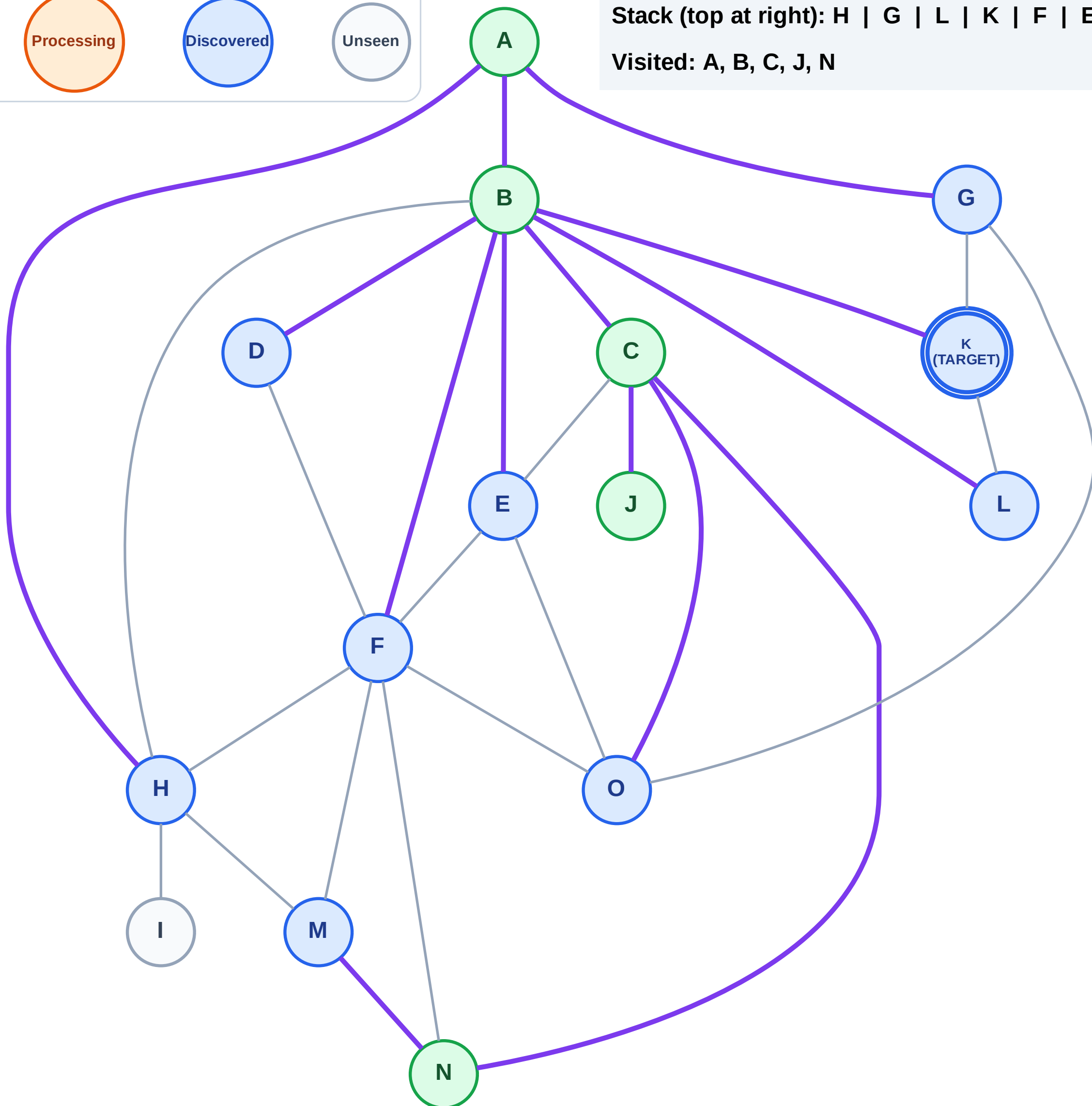
Step 11: Discover M from N

Legend



Stack (top at right): H | G | L | K | F | E | D | O | M

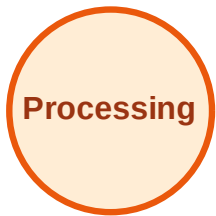
Visited: A, B, C, J, N



DFS Traversal

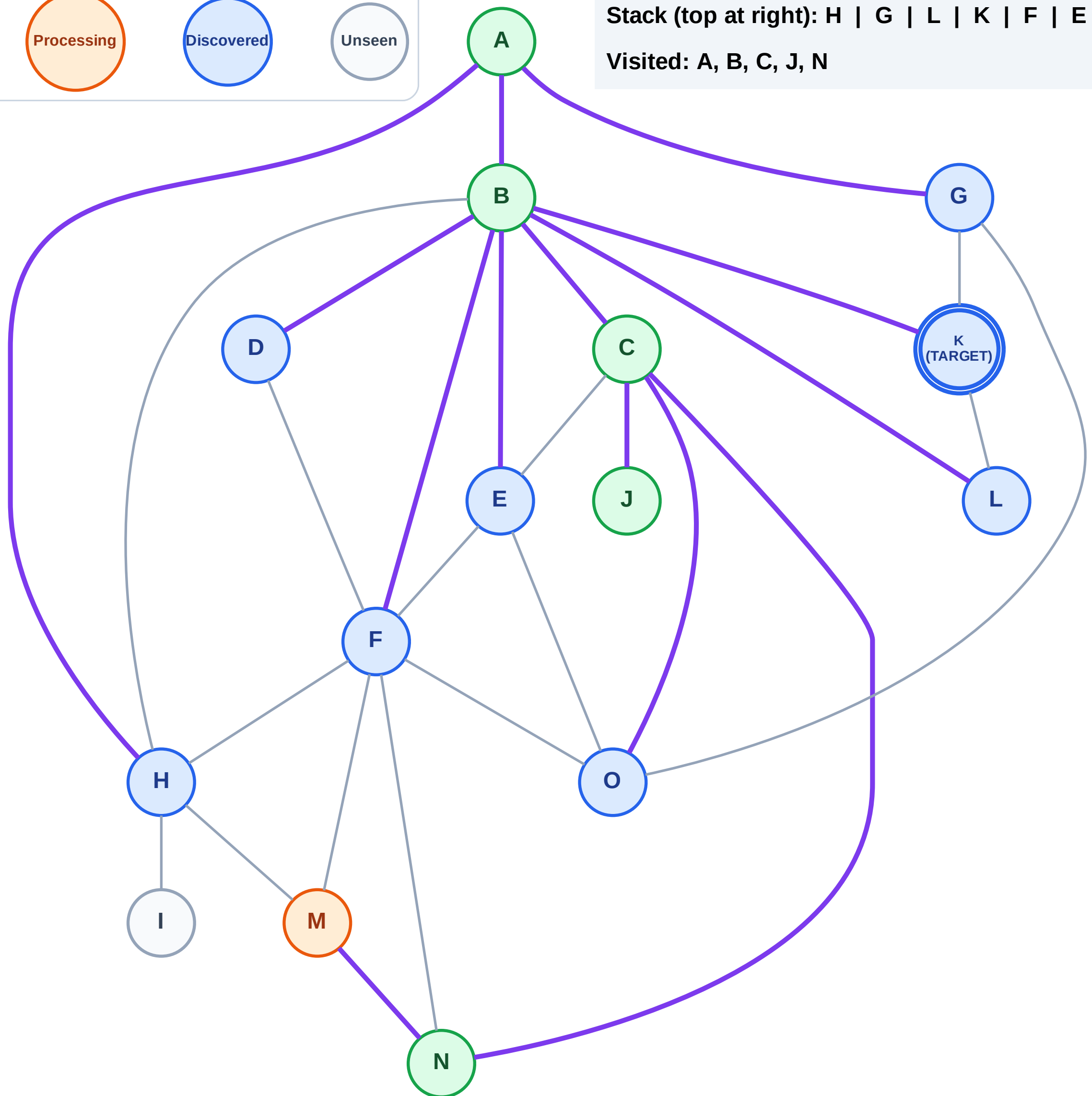
Step 12: Process node M

Legend



Stack (top at right): H | G | L | K | F | E | D | O

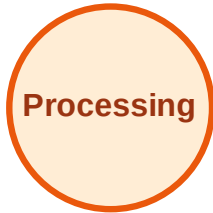
Visited: A, B, C, J, N



DFS Traversal

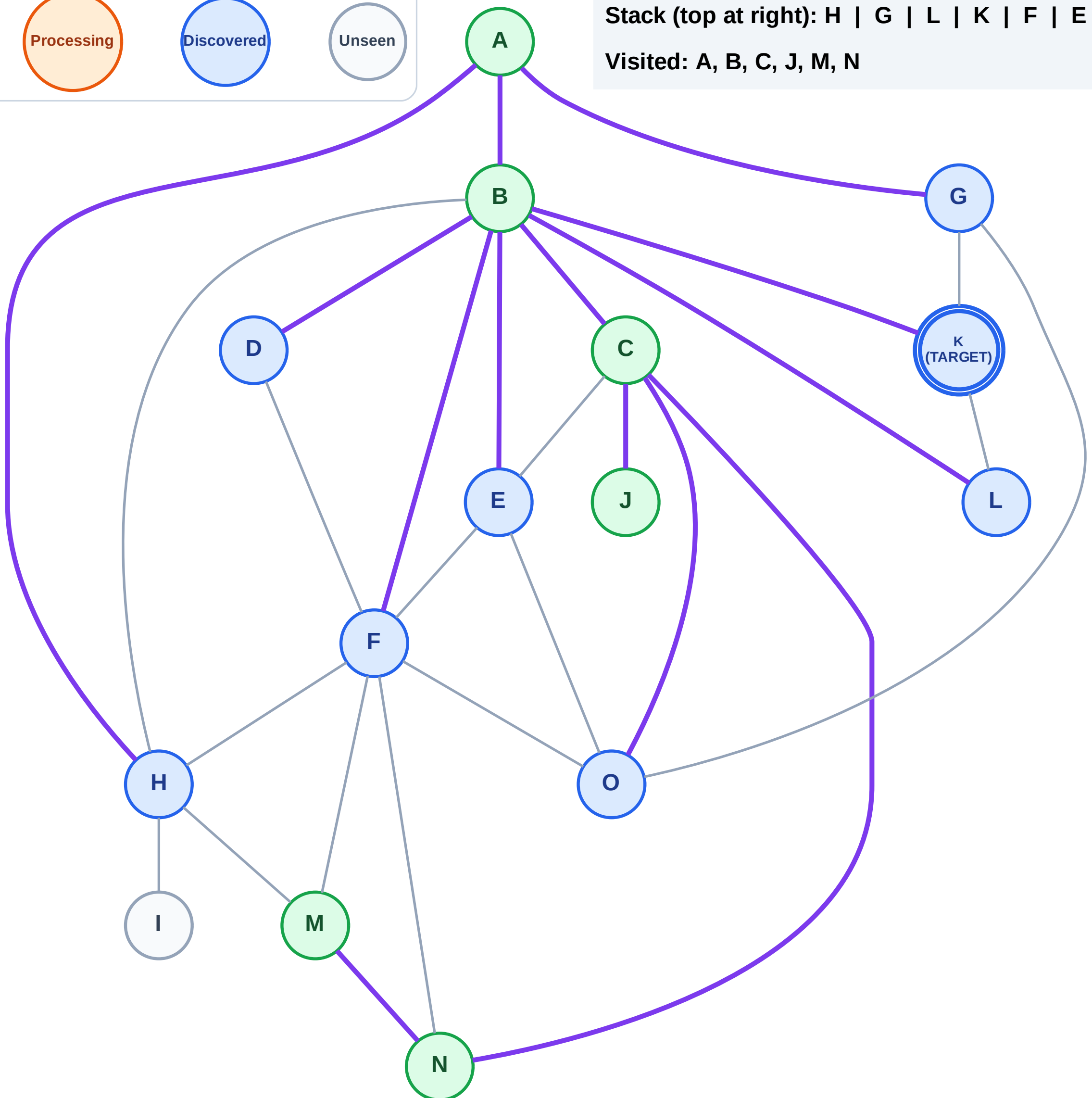
Step 13: No new neighbors from M

Legend



Stack (top at right): H | G | L | K | F | E | D | O

Visited: A, B, C, J, M, N



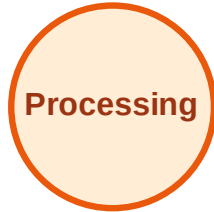
DFS Traversal

Step 14: Process node O

Legend



Complete



Processing



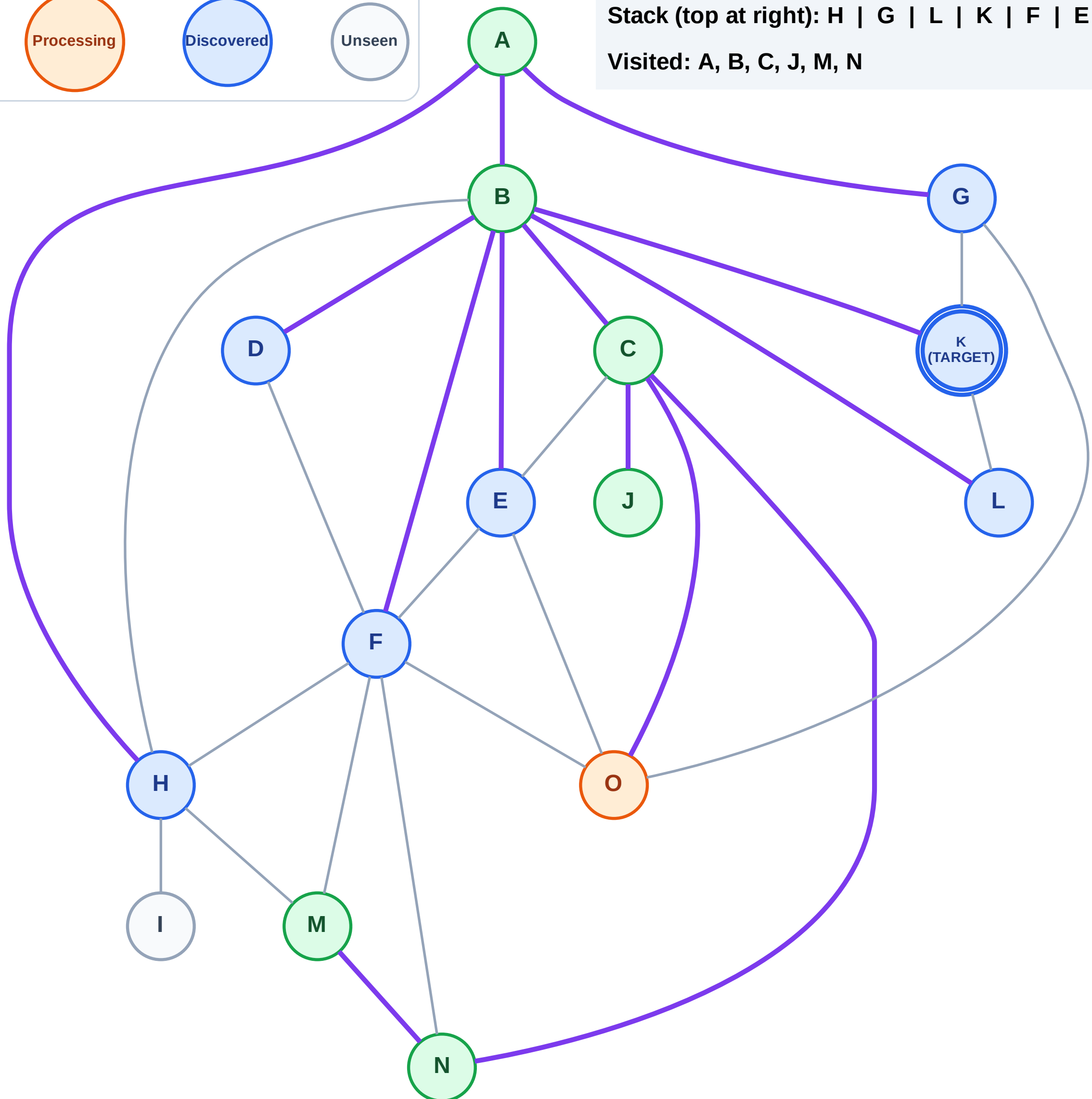
Discovered



Unseen

Stack (top at right): H | G | L | K | F | E | D

Visited: A, B, C, J, M, N



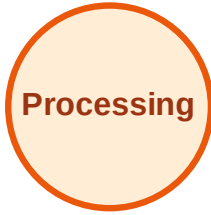
DFS Traversal

Step 15: No new neighbors from O

Legend



Complete



Processing



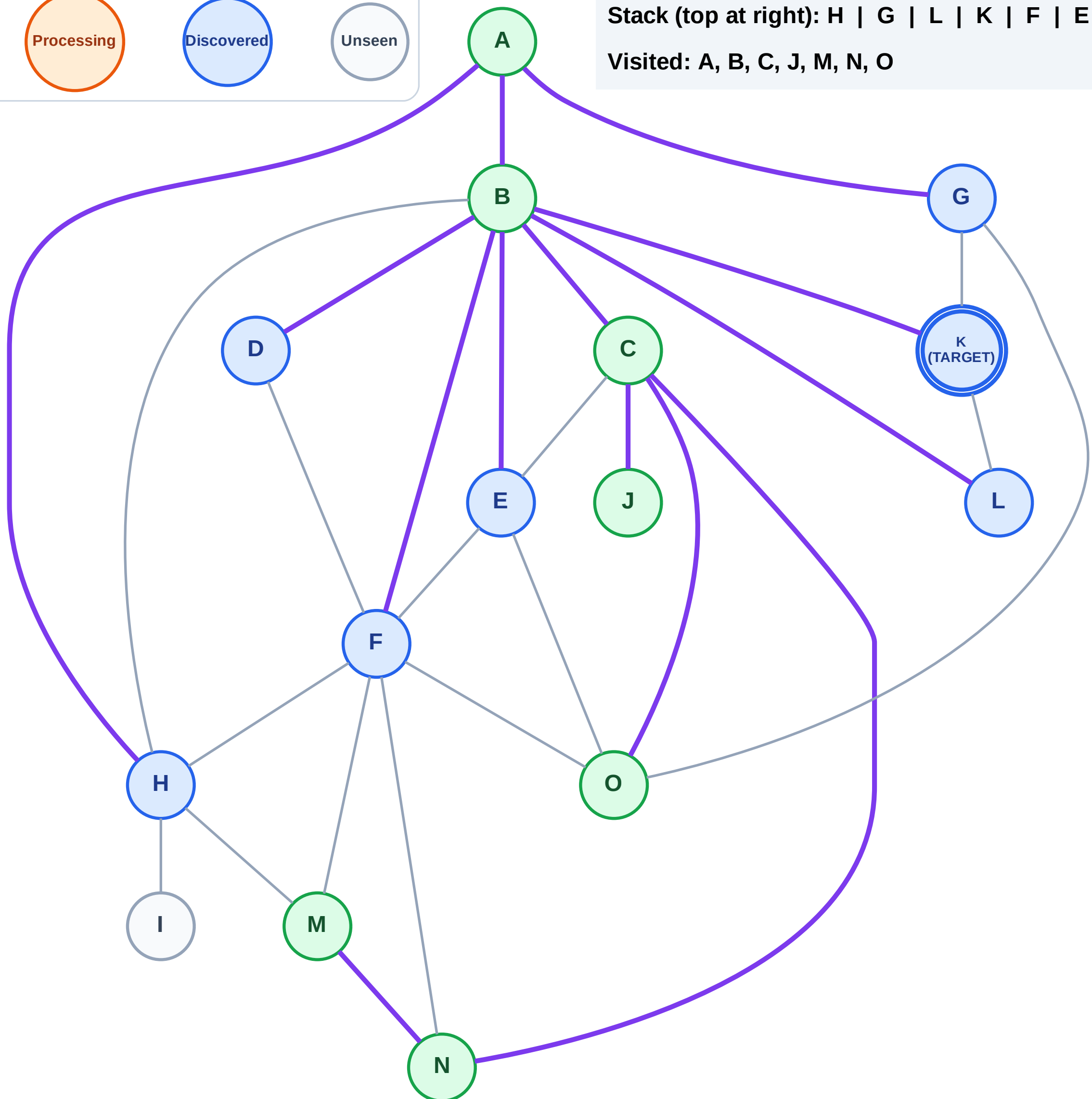
Discovered



Unseen

Stack (top at right): H | G | L | K | F | E | D

Visited: A, B, C, J, M, N, O



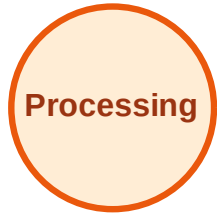
DFS Traversal

Step 16: Process node D

Legend



Complete



Processing



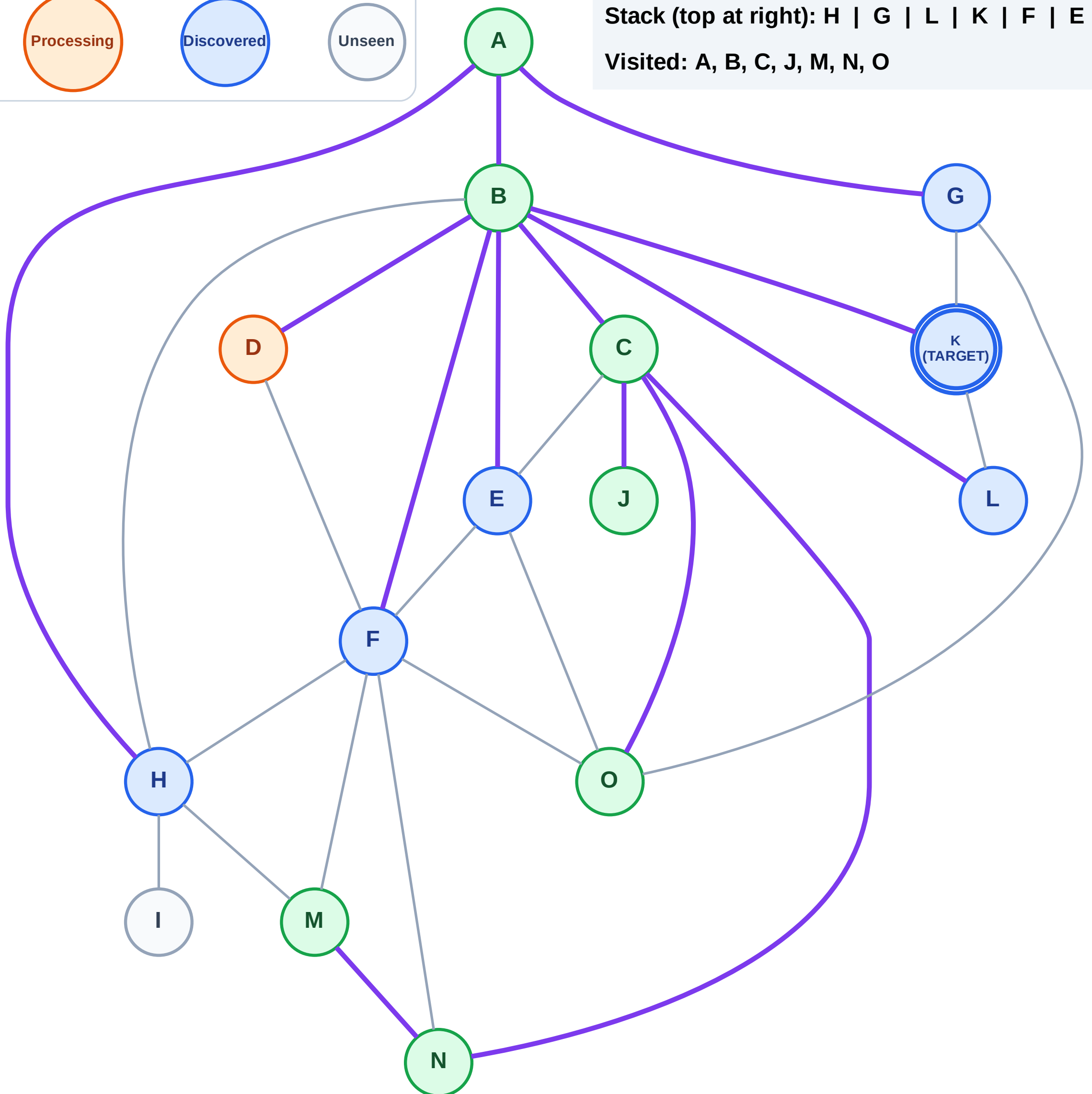
Discovered



Unseen

Stack (top at right): H | G | L | K | F | E

Visited: A, B, C, J, M, N, O



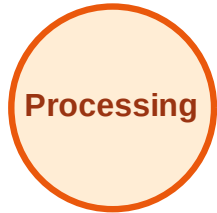
DFS Traversal

Step 17: No new neighbors from D

Legend



Complete



Processing



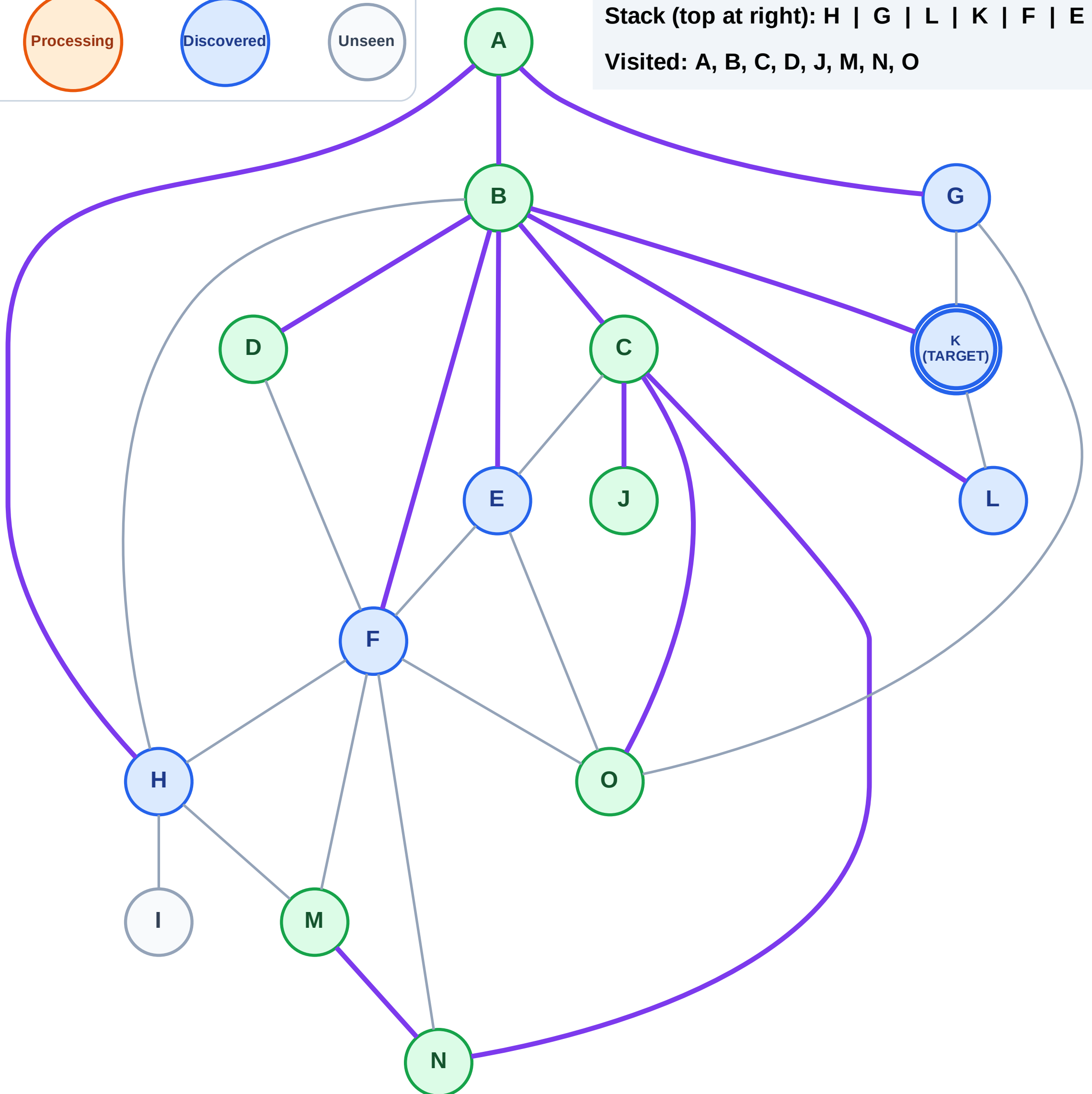
Discovered



Unseen

Stack (top at right): H | G | L | K | F | E

Visited: A, B, C, D, J, M, N, O



DFS Traversal

Step 18: Process node E

Legend

Complete

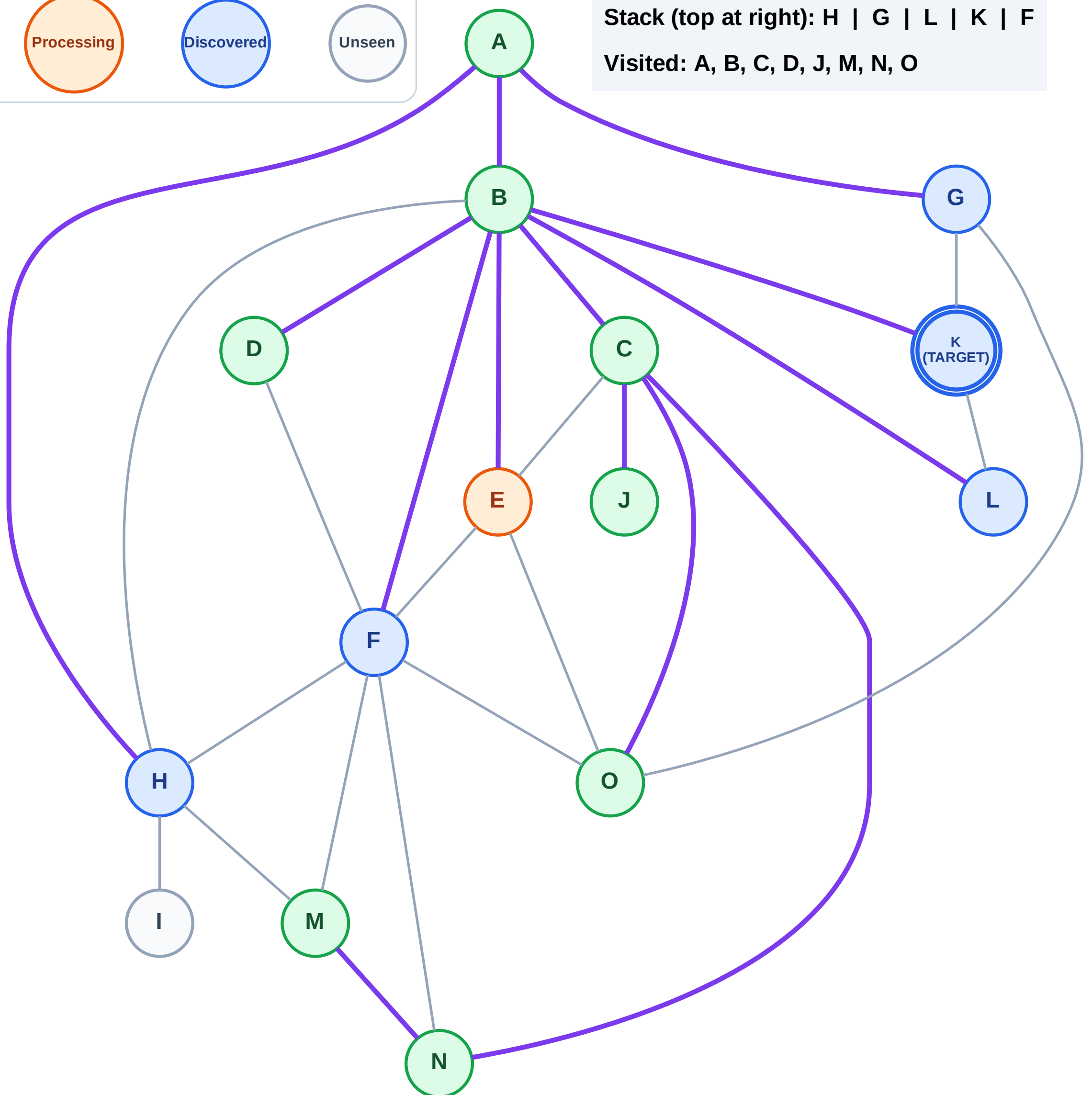
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K | F

Visited: A, B, C, D, J, M, N, O



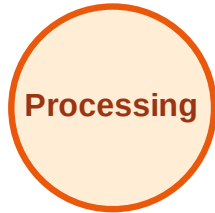
DFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



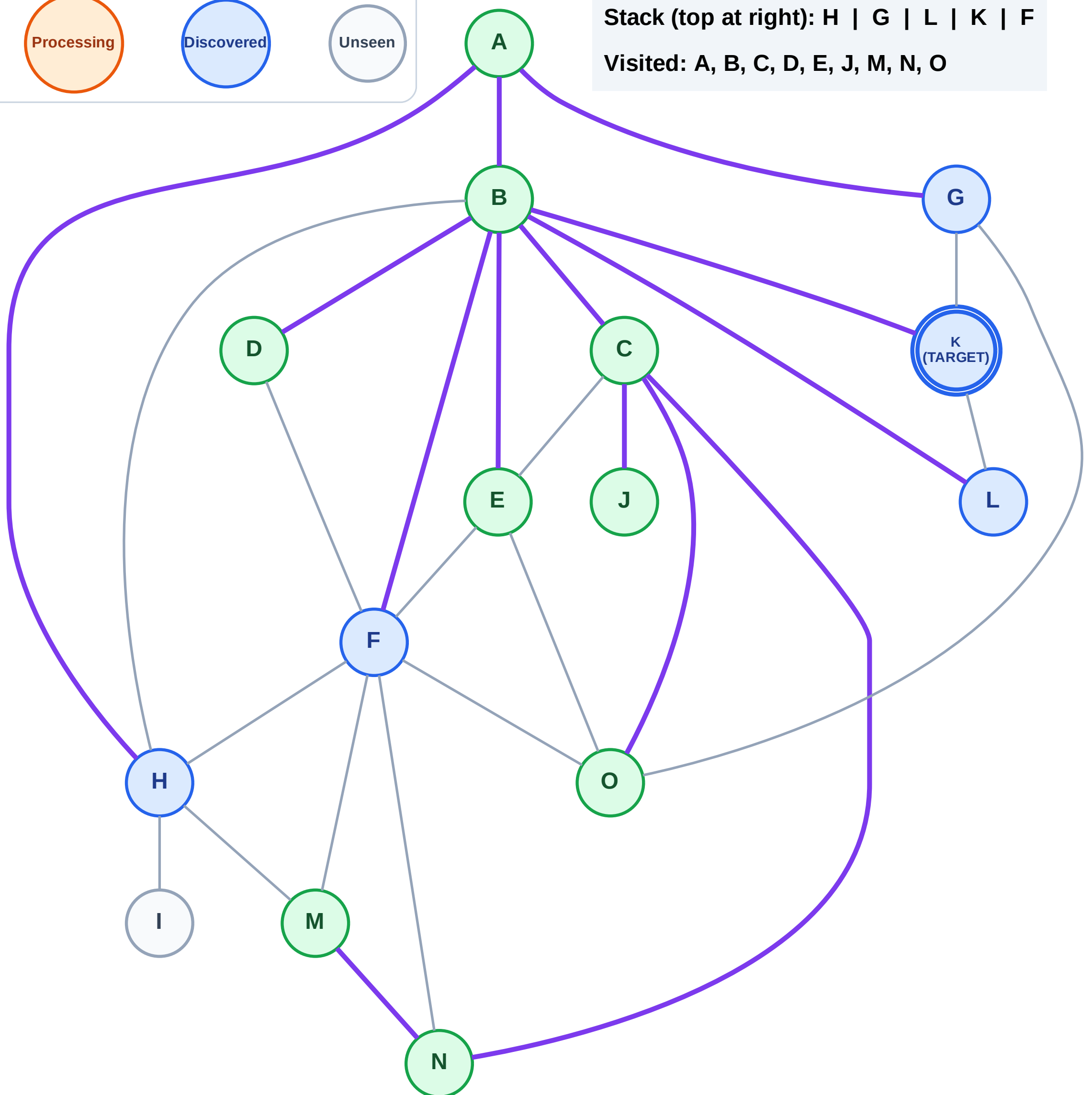
Discovered



Unseen

Stack (top at right): H | G | L | K | F

Visited: A, B, C, D, E, J, M, N, O



DFS Traversal

Step 20: Process node F

Legend

Complete

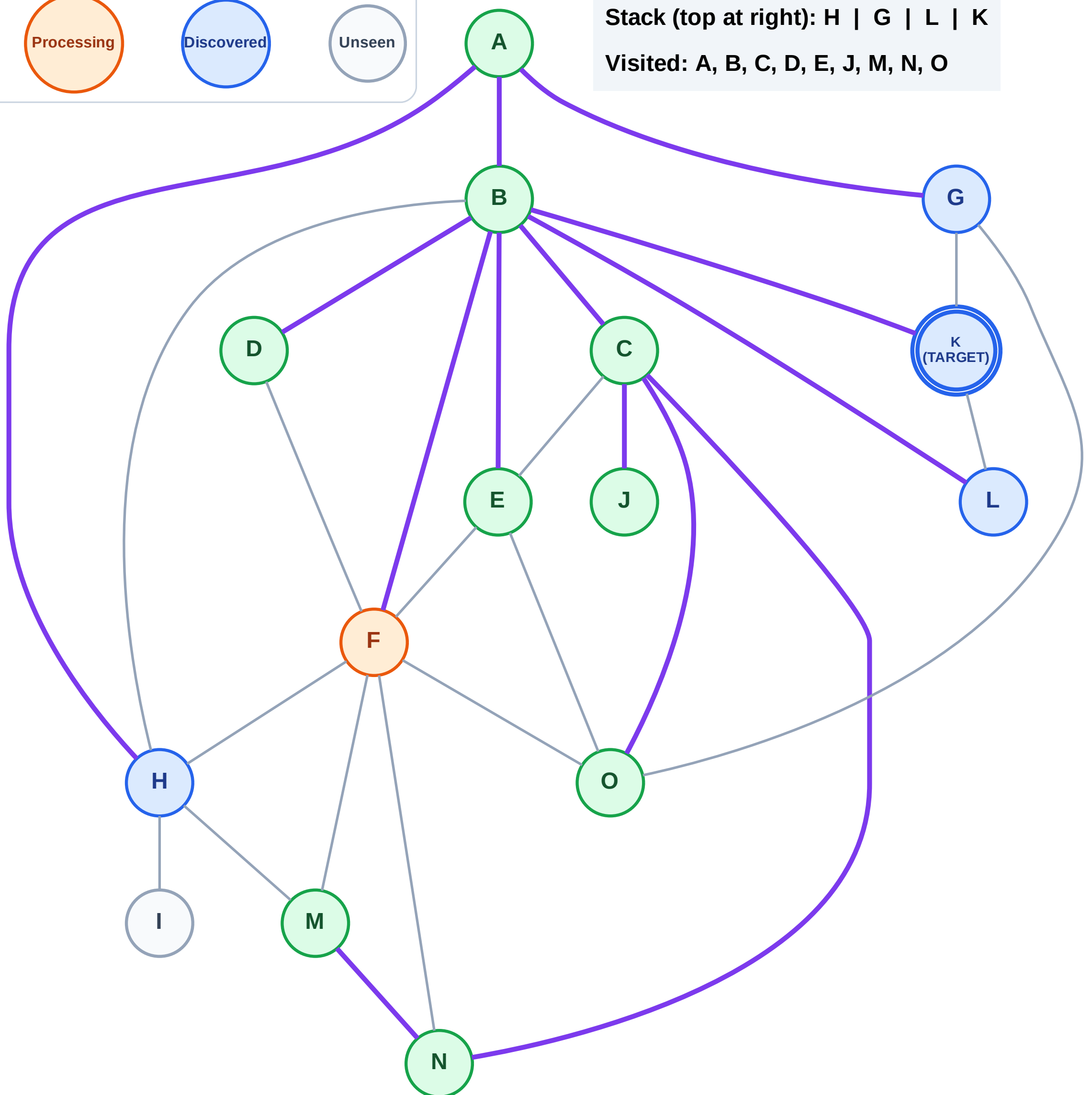
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K

Visited: A, B, C, D, E, J, M, N, O



DFS Traversal

Step 21: No new neighbors from F

Legend

Complete

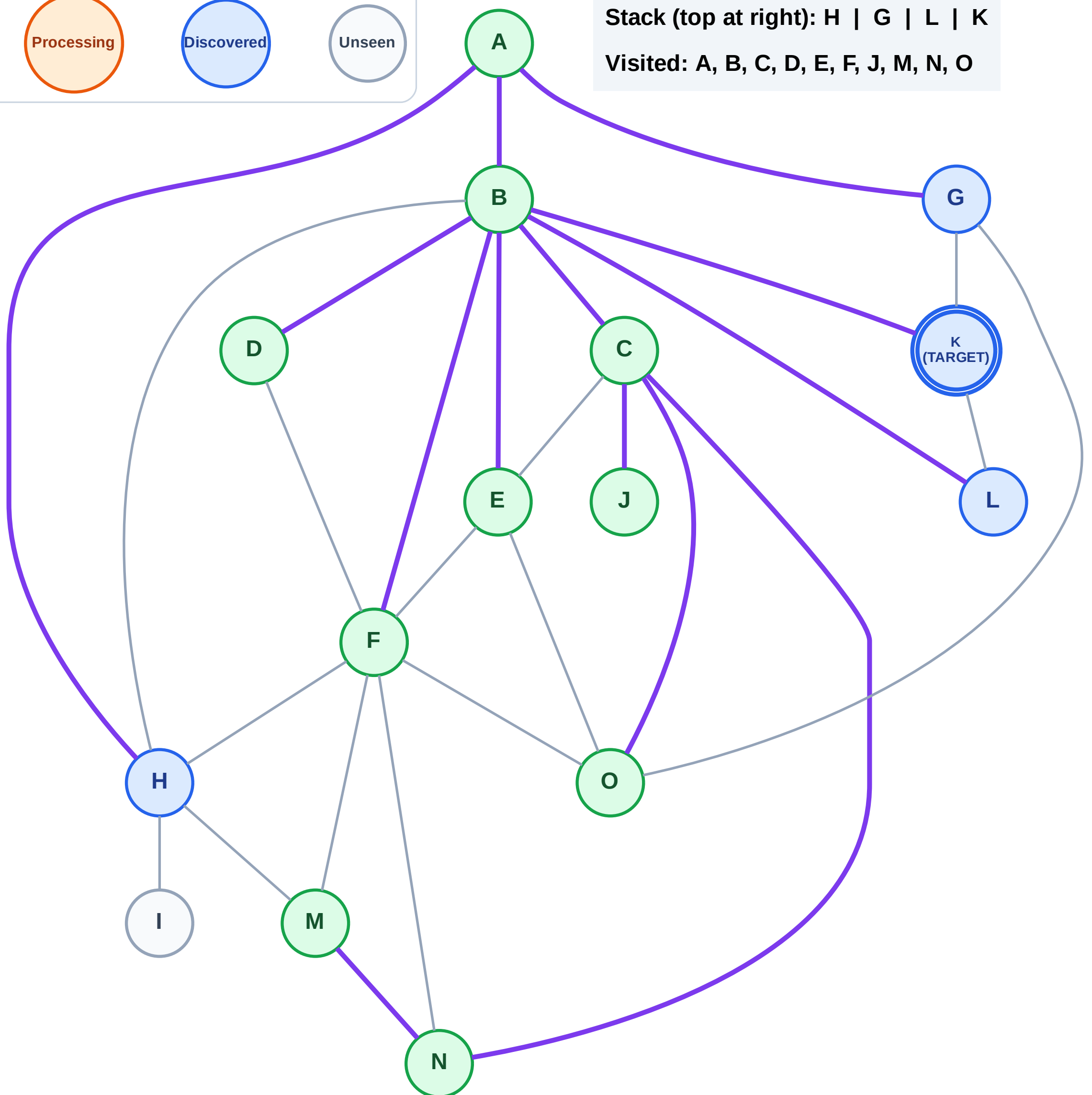
Processing

Discovered

Unseen

Stack (top at right): H | G | L | K

Visited: A, B, C, D, E, F, J, M, N, O



DFS Traversal

Step 22: Process node K

Legend

Complete

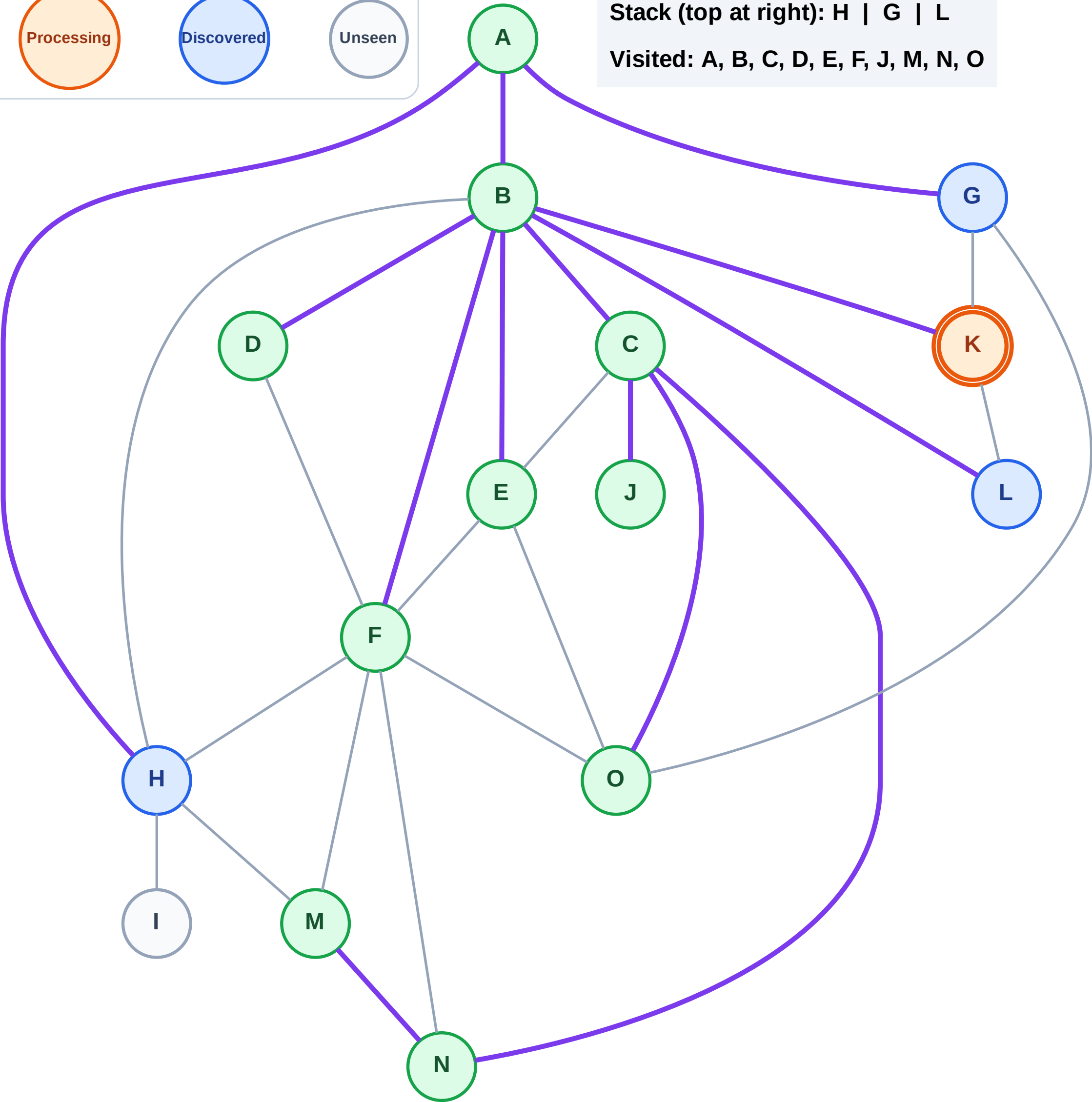
Processing

Discovered

Unseen

Stack (top at right): H | G | L

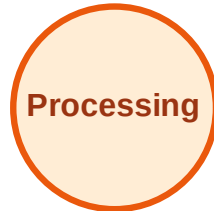
Visited: A, B, C, D, E, F, J, M, N, O



DFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): H | G | L

Visited: A, B, C, D, E, F, J, K, M, N, O

